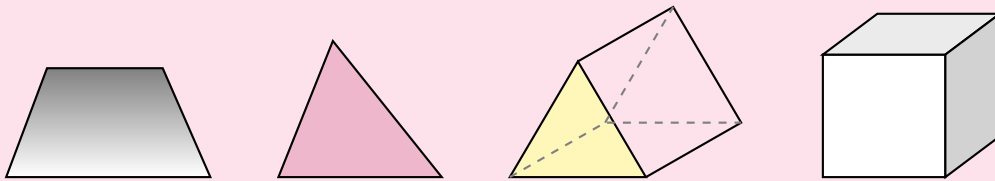

Unit 4

2D Shapes and 3D Solids

Revision Notes & Practice Worksheet



What is inside

Notes (pink pages)

- 4.1 Area of triangles, parallelograms, trapezia
- 4.2 Area of compound shapes
- 4.3 Properties of 3D solids & nets
- 4.4 Surface area
- 4.5 Volume
- 4.6 Measures of area and volume
- 4.7 Plans and elevations
- 4.8 Solving problems

Practice sets (white Q, yellow A)

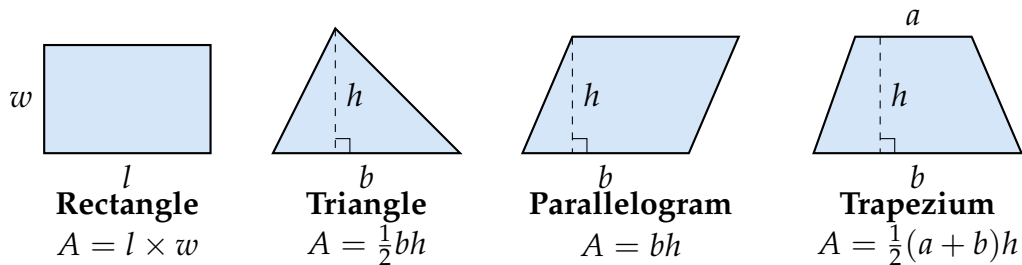
- A Area of triangles, parallelograms, trapezia
- B Compound shapes
- C 3D solids, faces, edges, vertices, nets
- D Surface area
- E Volume of cuboids and prisms
- F Converting units of area, volume, capacity
- G Plans and elevations
- H Mixed problem solving

How to use this worksheet. Read the pink notes first. Then try each white question page on your own, showing all working in the boxes. Only turn over to the yellow answer page when you have finished that page. Each topic has a **class practice** set (navy) and a **homework** set (teal), built from every question type in your reference sheets. Give units in every answer (cm, cm², cm³...). A calculator may be used.

Name: _____ Class: _____ Date: _____

Key Notes

4.1 Area of triangles, parallelograms and trapezia

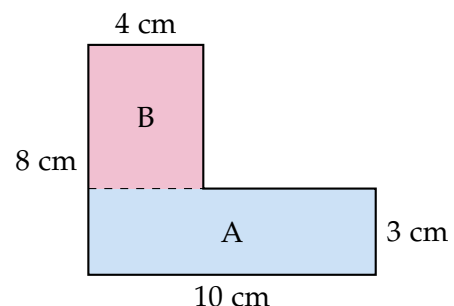


- h is always the **perpendicular height** – it meets the base at a right angle (90°). Never use a sloping (slant) side.
- A triangle is *half* of a parallelogram with the same base and height – that is why $A = \frac{1}{2}bh$.
- In a trapezium, a and b are the two **parallel** sides: add them, halve, multiply by the height.
- **Finding a missing length:** put the numbers into the formula, then solve.
 Triangle: $h = \frac{2A}{b}$ Parallelogram: $b = \frac{A}{h}$ Trapezium: $h = \frac{2A}{a + b}$
- Make sure all lengths are in the **same unit** before multiplying. Area units are squared: mm^2 , cm^2 , m^2 , km^2 .
- **Perimeter** = total distance round the outside (add every side). A square with perimeter P has side $P \div 4$. If two shapes have the *same perimeter*, write that as an equation.

4.2 Area of compound shapes

A **compound shape** is made from two or more simple shapes.

- **Add method:** split the shape into rectangles, triangles, trapezia..., find each area, then *add*.
- **Subtract method:** find the area of the big outer shape and *subtract* the missing part (a hole or a cut-out).
- Work out any **missing lengths** first using the lengths you know (e.g. $10 - 4 = 6$).
- **Perimeter** = distance all the way round (cm). **Area** = space inside (cm^2). Don't mix them up!



$$A = 10 \times 3 = 30, B = 4 \times 5 = 20$$

$$\text{Total} = 50 \text{ cm}^2$$

4.3 Properties of 3D solids

Face = flat (or curved) surface **Edge** = where two faces meet **Vertex** (plural *vertices*) = corner.

Solid	F	E	V	Solid	F	E	V
Cube / cuboid	6	12	8	Tetrahedron (triangular pyramid)	4	6	4
Triangular prism	5	9	6	Square-based pyramid	5	8	5
Pentagonal prism	7	15	10	Pentagonal pyramid	6	10	6
Hexagonal prism	8	18	12	Cylinder	3*	2	0

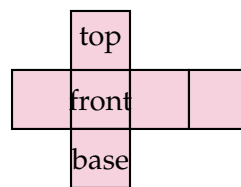
* 2 flat circular faces + 1 curved surface. Cone: 2 faces (1 flat, 1 curved), 1 edge, 1 vertex.
Sphere: 1 curved surface.

- A **prism** has the same **cross-section** all the way through. If the cross-section has n sides:
faces = $n + 2$, edges = $3n$, vertices = $2n$.
- A **pyramid** with an n -sided base: faces = $n + 1$, edges = $2n$, vertices = $n + 1$.
- **Euler's rule** (any solid with flat faces): $F + V - E = 2$
- Nets to recognise: **cylinder** = rectangle + 2 circles (one on each long edge); **cone** = sector (curved part) + circle; **square-based pyramid** = square + 4 triangles; **triangular prism** = 3 rectangles + 2 triangles.
- A **net** is a 2D shape that folds to make a 3D solid. A cube has 11 different nets – a net with 5 squares in a line or a 2×2 block is *not* a cube net. In a cube net, faces that are separated by exactly one square in a straight line are **opposite** each other.

4.4 Surface area

Surface area = the total area of all the faces of a 3D solid (think: area of its net). Units: cm^2 , m^2 .

- **Cube** (side l): $SA = 6l^2$
- **Cuboid**: $SA = 2(lw + lh + wh)$ (opposite faces are equal – find 3 faces and double)
- **Any prism**: $SA = 2 \times (\text{area of cross-section}) + (\text{perimeter of cross-section}) \times \text{length}$
- **Open box** (no lid): leave out the top face.
- **Pyramid**: base + each triangle ($\frac{1}{2} \times \text{base} \times \text{the triangle's own perpendicular height}$).
- **Composite solid**: count only the faces on the *outside* – leave out faces that are glued together.
- Sketch the net and label every face – then you won't miss one.

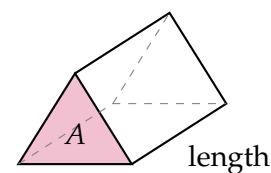


Net of a cube: 6 equal squares

4.5 Volume

Volume = the amount of 3D space a solid takes up. Units: mm^3 , cm^3 , m^3 .

- **Cuboid**: $V = l \times w \times h$ **Cube**: $V = l^3$
- **Prism**: $V = \text{area of cross-section} \times \text{length}$
- Find the cross-section area first (triangle, trapezium, L-shape...), then multiply by the length.
- **Rearranging**: $\text{length} = \frac{V}{\text{cross-section area}}$,
height of cuboid = $\frac{V}{l \times w}$, side of cube = $\sqrt[3]{V}$.
- Counting cubes: how many small cubes fit = (number along) $\times$ (number across) $\times$ (number up).



$$V = A \times \text{length}$$

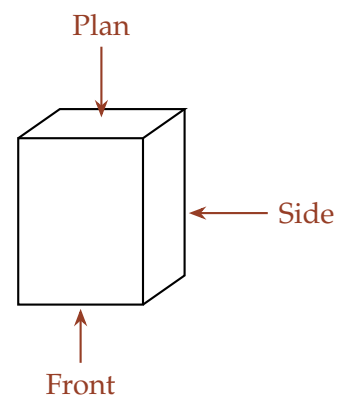
4.6 Measures of area and volume (converting units)

Length	Area (square the factor)	Volume (cube the factor)
1 cm = 10 mm	$1 \text{ cm}^2 = 10^2 = 100 \text{ mm}^2$	$1 \text{ cm}^3 = 10^3 = 1000 \text{ mm}^3$
1 m = 100 cm	$1 \text{ m}^2 = 100^2 = 10\,000 \text{ cm}^2$	$1 \text{ m}^3 = 100^3 = 1\,000\,000 \text{ cm}^3$
1 km = 1000 m	$1 \text{ km}^2 = 1\,000\,000 \text{ m}^2$	

- Big unit → small unit: **multiply**. Small unit → big unit: **divide**.
- **Hectare**: $1 \text{ ha} = 10\,000 \text{ m}^2$ (a square 100 m by 100 m). $1 \text{ km}^2 = 100 \text{ ha}$.
- **Capacity**: $1 \text{ cm}^3 = 1 \text{ ml}$ $1000 \text{ cm}^3 = 1 \text{ litre}$ $1 \text{ m}^3 = 1000 \text{ litres}$ $1 \text{ litre} = 1000 \text{ ml}$.
- **Mass**: $1 \text{ tonne (t)} = 1000 \text{ kg}$ $1 \text{ kg} = 1000 \text{ g}$.
- Tip: it is often easier to **convert the lengths first**, then work out the area or volume.

4.7 Plans and elevations

- **Plan** = the view from directly **above**.
- **Front elevation** = the view from the **front**.
- **Side elevation** = the view from the **side**.
- Each view is a flat 2D shape. Draw accurately with a ruler and label the lengths.
- Plan of a cylinder or cone = a circle (cone: circle with a dot at the centre). Plan of a square-based pyramid = a square with its diagonals.
- **Isometric paper** (dots in triangles) is used to draw 3D solids; each gap between dots = 1 unit. Vertical edges stay vertical.
- **Scale**: “2 cm to 1 m” means real metres $\times 2 =$ cm on the drawing.
- Working backwards: front gives width & height, side gives depth & height, plan gives width & depth.



4.8 Solving problems with 3D solids and measures

1. Decide what is being asked: **length** (cm), **area / surface area** (cm²) or **volume / capacity** (cm³, litres)?
2. Draw a sketch and label it. Put all lengths in the same unit.
3. Choose the formula, substitute, calculate – show every step.
4. Answer the actual question (e.g. “how many tins?” – round **up** if you need enough; “how many fit?” – round **down**).
5. Check: does the answer make sense? Have you written the units?
6. **Rates:** amount per minute = cross-section area $\times$ speed; time = total volume $\div$ rate. Convert hours/minutes/seconds carefully.

Common mistakes to avoid

- Using the **slant side** instead of the perpendicular height.
- Forgetting the $\frac{1}{2}$ for triangles and trapezia.
- Thinking $1 \text{ m}^2 = 100 \text{ cm}^2$ (it is $10\,000 \text{ cm}^2$) or $1 \text{ cm}^3 = 10 \text{ mm}^3$ (it is 1000 mm^3).
- Missing faces when finding surface area (a cuboid has 6, a triangular prism has 5).
- Writing cm² for a volume or cm³ for an area.

A Area of triangles, parallelograms & trapezia

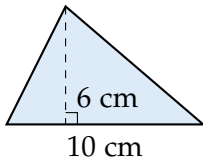
Section 4.1

1.

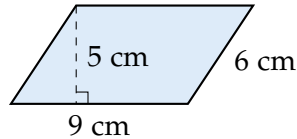
[Maximum mark: 8]

Work out the area of each shape. (*Diagrams not drawn to scale.*)

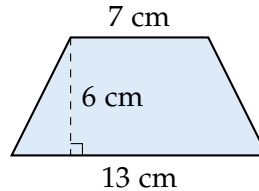
a



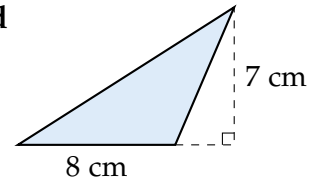
b



c



d



2.

[Maximum mark: 6]

Find the missing length in each shape.

- a) A triangle has area 36 cm^2 and base 9 cm. Find its perpendicular height.
- b) A parallelogram has area 84 cm^2 and perpendicular height 7 cm. Find its base.
- c) A trapezium has area 50 cm^2 . Its parallel sides are 6 cm and 14 cm. Find its height.

3.

[Maximum mark: 2]

A triangle has base 40 mm and perpendicular height 3 cm. Work out its area in cm^2 .

4.

[Maximum mark: 3]

A trapezium has area 72 cm^2 and height 8 cm. One of its parallel sides is 5 cm long. How long is the other parallel side?

Answers – A Area of 2D shapes

worked solutions

1. a) $A = \frac{1}{2} \times 10 \times 6 = \boxed{30 \text{ cm}^2}$ b) $A = bh = 9 \times 5 = \boxed{45 \text{ cm}^2}$ (the 6 cm slant side is not used)

c) $A = \frac{1}{2}(7 + 13) \times 6 = \frac{1}{2} \times 20 \times 6 = \boxed{60 \text{ cm}^2}$ d) $A = \frac{1}{2} \times 8 \times 7 = \boxed{28 \text{ cm}^2}$ (height can be outside the triangle)

2. a) $36 = \frac{1}{2} \times 9 \times h \Rightarrow 36 = 4.5h \Rightarrow h = \boxed{8 \text{ cm}}$

b) $84 = b \times 7 \Rightarrow b = 84 \div 7 = \boxed{12 \text{ cm}}$

c) $50 = \frac{1}{2}(6 + 14) \times h = 10h \Rightarrow h = \boxed{5 \text{ cm}}$

3. Same units first: 40 mm = 4 cm. $A = \frac{1}{2} \times 4 \times 3 = \boxed{6 \text{ cm}^2}$

4. $72 = \frac{1}{2}(a + 5) \times 8 = 4(a + 5) \Rightarrow a + 5 = 18 \Rightarrow a = \boxed{13 \text{ cm}}$

A Area of 2D shapes – part 2

Section 4.1

5.

[Maximum mark: 3]

A parallelogram has base 12 cm and perpendicular height 5 cm.

A triangle with base 15 cm has the **same area** as the parallelogram. Find the perpendicular height of the triangle.

6.

[Maximum mark: 4]

A flowerbed is in the shape of a trapezium. Its parallel sides are 4.5 m and 7.5 m and they are 4 m apart.

One box of grass seed covers 10 m^2 and costs £3.50. How much will it cost to cover the flowerbed?

7.

[Maximum mark: 2]

A parallelogram has base 8 cm, sloping side 5 cm and perpendicular height 4 cm.

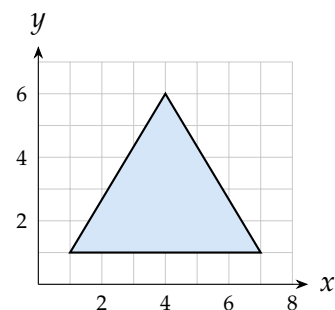
Sam says, "The area is $8 \times 5 = 40 \text{ cm}^2$." Explain Sam's mistake and give the correct area.

8.

[Maximum mark: 3]

A triangle has vertices at (1,1), (7,1) and (4,6), as shown on the grid.

Work out the area of the triangle.



Answers – A Area of 2D shapes – part 2 worked solutions

5. Parallelogram: $12 \times 5 = 60 \text{ cm}^2$. Triangle: $\frac{1}{2} \times 15 \times h = 60 \Rightarrow 7.5h = 60 \Rightarrow h = \boxed{8 \text{ cm}}$

6. Area = $\frac{1}{2}(4.5 + 7.5) \times 4 = \frac{1}{2} \times 12 \times 4 = 24 \text{ m}^2$.

Boxes: $24 \div 10 = 2.4$, so **3 boxes** are needed (round up). Cost = $3 \times \text{£}3.50 = \boxed{\text{£}10.50}$

7. Sam multiplied the base by the **sloping side**. The area of a parallelogram uses the **perpendicular height**:

$$A = 8 \times 4 = \boxed{32 \text{ cm}^2}$$

8. Base from $(1, 1)$ to $(7, 1) = 6$ units. Height from $y = 1$ up to $y = 6 = 5$ units.

$$A = \frac{1}{2} \times 6 \times 5 = \boxed{15 \text{ square units}}$$

Homework A Area & perimeter of 2D shapes Section 4.1

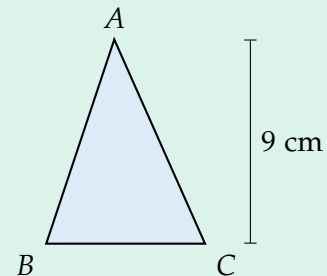
Diagrams are not drawn to scale.

1.

[Maximum mark: 2]

Triangle ABC has a perpendicular height of 9 cm and an area of 63 cm^2 .

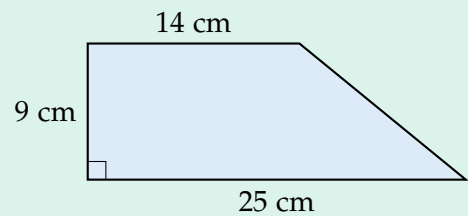
Work out the length of BC .



[Maximum mark: 2]

2.

Find the area of this trapezium.

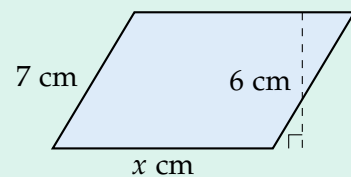


[Maximum mark: 2]

3.

The area of this parallelogram is 58.2 cm^2 .

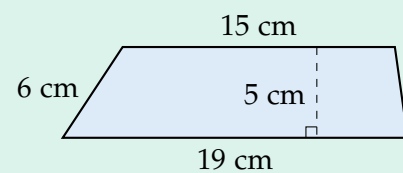
Work out the value of x .



[Maximum mark: 2]

4.

Calculate the area of this trapezium.



[Maximum mark: 2]

5.

The area of a triangle is 621 cm^2 . The length of its base is 27 cm.

Calculate the perpendicular height of the triangle.

Homework answers A Area & perimeter of 2D shapes

1. $63 = \frac{1}{2} \times BC \times 9 \Rightarrow 63 = 4.5 \times BC \Rightarrow BC = 63 \div 4.5 = \boxed{14 \text{ cm}}$
2. $A = \frac{1}{2}(14 + 25) \times 9 = \frac{1}{2} \times 39 \times 9 = \boxed{175.5 \text{ cm}^2}$
3. $A = bh \Rightarrow 58.2 = x \times 6 \Rightarrow x = 58.2 \div 6 = \boxed{9.7}$ (the 7 cm sloping side is not needed)
4. $A = \frac{1}{2}(15 + 19) \times 5 = \frac{1}{2} \times 34 \times 5 = \boxed{85 \text{ cm}^2}$ (6 cm is the sloping side – not used)
5. $621 = \frac{1}{2} \times 27 \times h \Rightarrow 621 = 13.5h \Rightarrow h = 621 \div 13.5 = \boxed{46 \text{ cm}}$

Homework A Area & perimeter – part 2

Section 4.1

6.

[Maximum mark: 3]

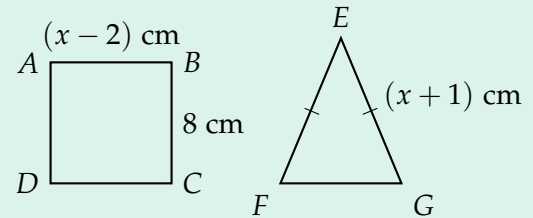
$ABCD$ is a square with $AB = (x - 2)$ cm and $BC = 8$ cm.

a) Find the value of x .

Triangle EFG is isosceles with $EF = EG = (x + 1)$ cm.

The square and the triangle have the **same perimeter**.

b) Work out the length of FG .



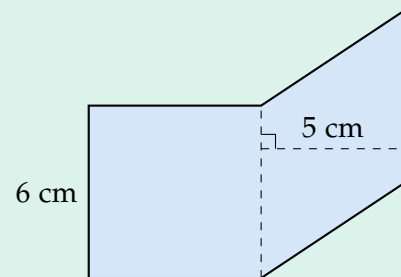
7.

[Maximum mark: 3]

The shaded shape is made by joining a square and a rhombus. The sides of both are 6 cm. The perpendicular distance between the parallel vertical sides of the rhombus is 5 cm.

Work out

a) the perimeter of the shaded shape b) its area.



8.

[Maximum mark: 3]

A rectangle measures 9 cm by 5 cm. A square has the same perimeter as the rectangle. Work out the area of the square.

Homework answers A Area & perimeter – part 2

6. a) A square has equal sides, so $x - 2 = 8 \Rightarrow x = \boxed{10}$

b) Perimeter of square = $4 \times 8 = 32$ cm. $EF = EG = 10 + 1 = 11$ cm.

$$FG = 32 - 11 - 11 = \boxed{10 \text{ cm}}$$

7. a) The outside edge is 3 sides of the square and 3 sides of the rhombus (the shared side is inside):

$$6 \times 6 = \boxed{36 \text{ cm}}$$

b) Square = $6 \times 6 = 36 \text{ cm}^2$. Rhombus (a parallelogram) = base $\times$ height = $6 \times 5 = 30 \text{ cm}^2$.

$$\text{Total} = \boxed{66 \text{ cm}^2}$$

8. Rectangle perimeter = $9 + 5 + 9 + 5 = 28$ cm. Square side = $28 \div 4 = 7$ cm. Area = $7 \times 7 = \boxed{49 \text{ cm}^2}$

B Area of compound shapes

Section 4.2

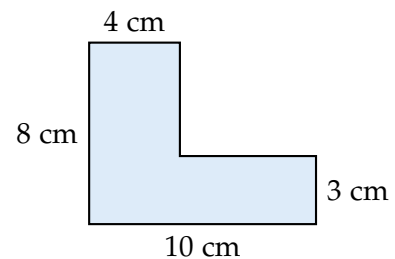
All diagrams are made of straight lines and right angles unless shown. Not drawn to scale.

1.

[Maximum mark: 5]

For this L-shape, work out

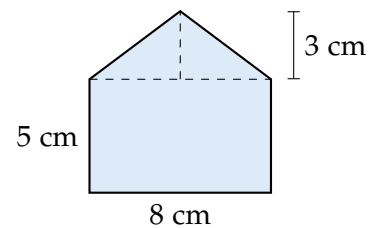
a) the area b) the perimeter.



2.

[Maximum mark: 3]

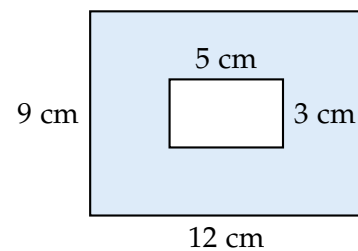
The shape is a rectangle with a triangle on top.
Work out its total area.



3.

[Maximum mark: 3]

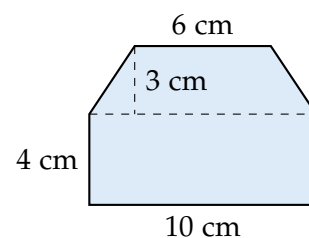
A rectangular photo frame is 12 cm by 9 cm. The rectangular hole in the middle is 5 cm by 3 cm.
Work out the area of the shaded frame.



4.

[Maximum mark: 4]

The shape is made from a rectangle and a trapezium.
Work out its total area.



Answers – B Compound shapes

worked solutions

1. a) Split into two rectangles: bottom $10 \times 3 = 30$; top part $4 \times (8 - 3) = 4 \times 5 = 20$.

$$\text{Area} = 30 + 20 = \boxed{50 \text{ cm}^2}$$

b) Missing sides: $10 - 4 = 6 \text{ cm}$ and $8 - 3 = 5 \text{ cm}$.

$$\text{Perimeter} = 10 + 3 + 6 + 5 + 4 + 8 = \boxed{36 \text{ cm}}$$

2. Rectangle $= 8 \times 5 = 40 \text{ cm}^2$. Triangle $= \frac{1}{2} \times 8 \times 3 = 12 \text{ cm}^2$. Total $= \boxed{52 \text{ cm}^2}$

3. Subtract method: $12 \times 9 - 5 \times 3 = 108 - 15 = \boxed{93 \text{ cm}^2}$

4. Rectangle $= 10 \times 4 = 40 \text{ cm}^2$. Trapezium $= \frac{1}{2}(6 + 10) \times 3 = 24 \text{ cm}^2$. Total $= \boxed{64 \text{ cm}^2}$

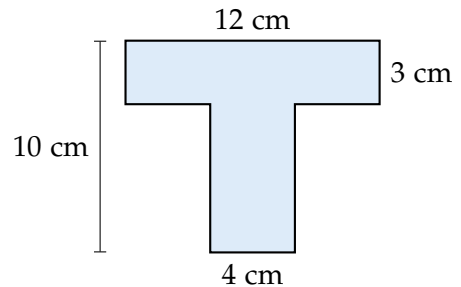
B Area of compound shapes – part 2

Section 4.2

5.

[Maximum mark: 3]

Work out the area of this T-shape.



6.

[Maximum mark: 3]

A rectangular garden measures 20 m by 15 m. A lawn in the middle is surrounded by a path 2 m wide all the way round. Work out the area of the path.

7.

[Maximum mark: 5]

A wall is 5 m long and 2.4 m high. It has a door 0.9 m by 2 m and a window 1.2 m by 1 m.

Ali paints the wall (not the door or window) with **two coats**. One litre of paint covers 6 m². Paint is sold in 2.5 litre tins.

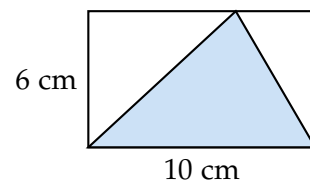
How many tins does Ali need to buy?

8.

[Maximum mark: 3]

The rectangle is 10 cm by 6 cm. The shaded triangle has its base along the bottom of the rectangle and its top vertex on the top edge.

Work out the total area that is **not** shaded.



Answers – B Compound shapes – part 2 worked solutions

5. Top bar = $12 \times 3 = 36 \text{ cm}^2$. Stem: height = $10 - 3 = 7 \text{ cm}$, so $4 \times 7 = 28 \text{ cm}^2$.

$$\text{Total} = 36 + 28 = \boxed{64 \text{ cm}^2}$$

6. Garden = $20 \times 15 = 300 \text{ m}^2$. Lawn = $(20 - 2 - 2) \times (15 - 2 - 2) = 16 \times 11 = 176 \text{ m}^2$.

Path = $300 - 176 = \boxed{124 \text{ m}^2}$ (Remember the path is on *both* sides, so subtract 4 m from each length.)

7. Wall = $5 \times 2.4 = 12 \text{ m}^2$. Door = $0.9 \times 2 = 1.8 \text{ m}^2$. Window = $1.2 \times 1 = 1.2 \text{ m}^2$.

Area to paint = $12 - 1.8 - 1.2 = 9 \text{ m}^2$. Two coats = 18 m^2 .

Paint needed = $18 \div 6 = 3$ litres. Tins: $3 \div 2.5 = 1.2$, so Ali needs $\boxed{2 \text{ tins}}$ (round up).

8. Rectangle = $10 \times 6 = 60 \text{ cm}^2$. Shaded triangle = $\frac{1}{2} \times 10 \times 6 = 30 \text{ cm}^2$.

Not shaded = $60 - 30 = \boxed{30 \text{ cm}^2}$ (It doesn't matter where the top vertex is!)

Homework B Area of compound shapes

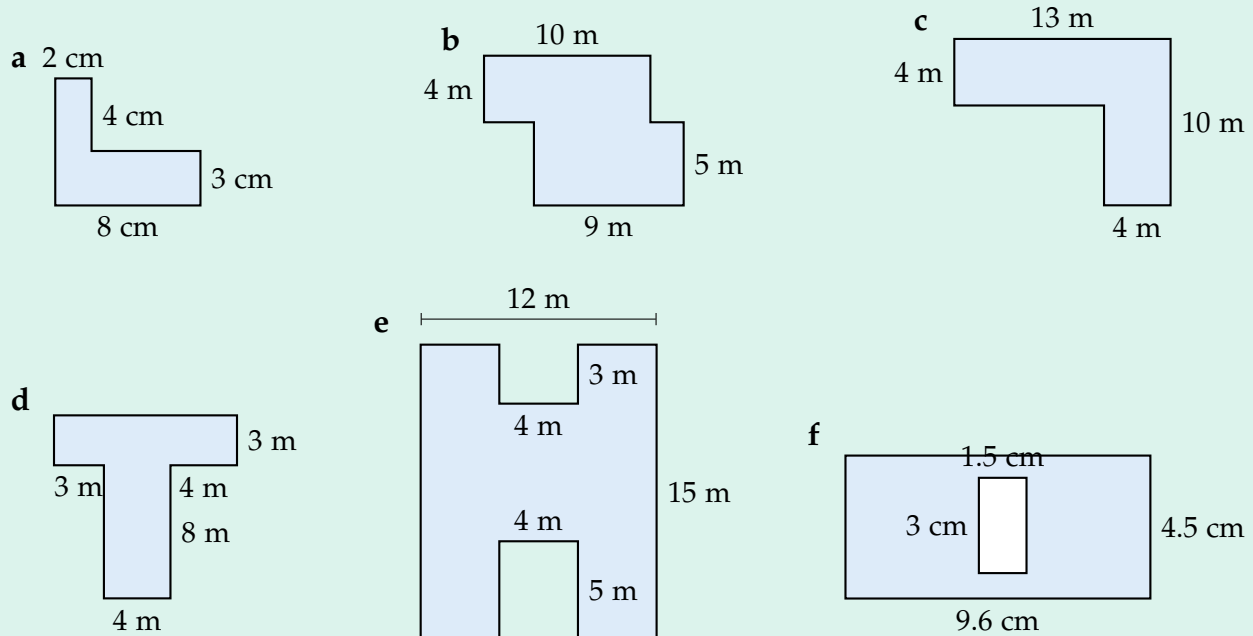
Section 4.2

All corners are right angles unless shown. Diagrams are not drawn to scale.

1.

[Maximum mark: 12]

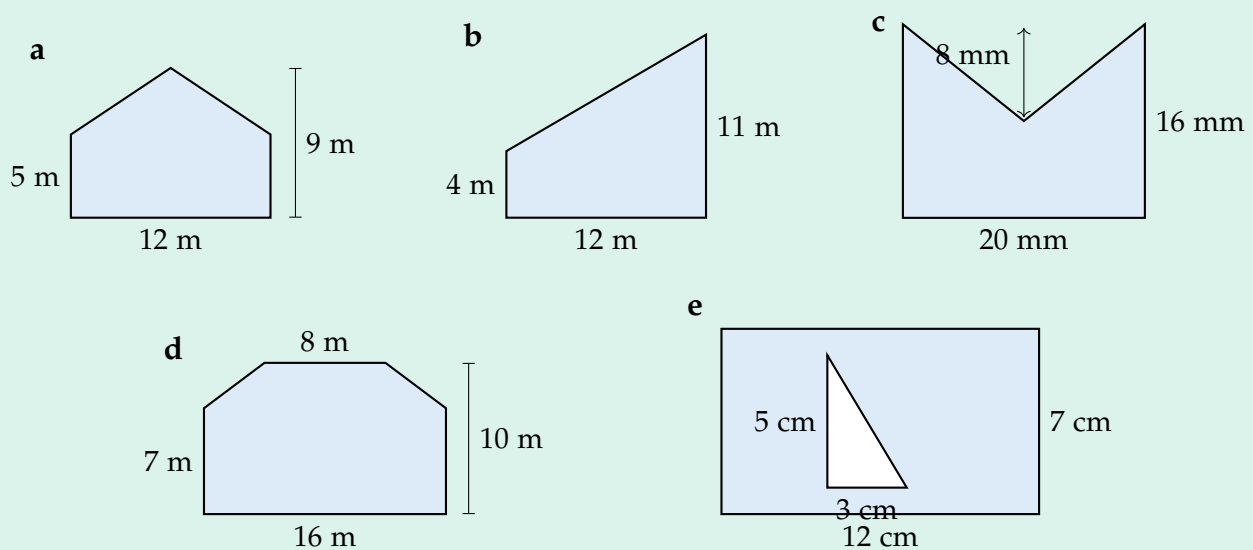
Work out the area of each shaded shape.



2.

[Maximum mark: 10]

Work out the area of each shaded shape.



Homework answers B Compound shapes

1. a) $8 \times 3 + 2 \times 4 = 24 + 8 = 32 \text{ cm}^2$ b) $10 \times 4 + 9 \times 5 = 40 + 45 = 85 \text{ m}^2$

c) $13 \times 4 + 4 \times 6 = 52 + 24 = 76 \text{ m}^2$ (right part height = $10 - 4 = 6$)

d) Bar: $(3 + 4 + 4) \times 3 = 33$; stem: $4 \times 8 = 32$. Total = 65 m^2

e) $12 \times 15 - 4 \times 3 - 4 \times 5 = 180 - 12 - 20 = 148 \text{ m}^2$

f) $9.6 \times 4.5 - 1.5 \times 3 = 43.2 - 4.5 = 38.7 \text{ cm}^2$

2. a) Rectangle $12 \times 5 = 60$; triangle $\frac{1}{2} \times 12 \times (9 - 5) = 24$. Total = 84 m^2

b) It is a trapezium: $\frac{1}{2}(4 + 11) \times 12 = 90 \text{ m}^2$

c) $20 \times 16 - \frac{1}{2} \times 20 \times 8 = 320 - 80 = 240 \text{ mm}^2$

d) Rectangle $16 \times 7 = 112$; trapezium $\frac{1}{2}(8 + 16) \times (10 - 7) = 36$. Total = 148 m^2

e) $12 \times 7 - \frac{1}{2} \times 3 \times 5 = 84 - 7.5 = 76.5 \text{ cm}^2$

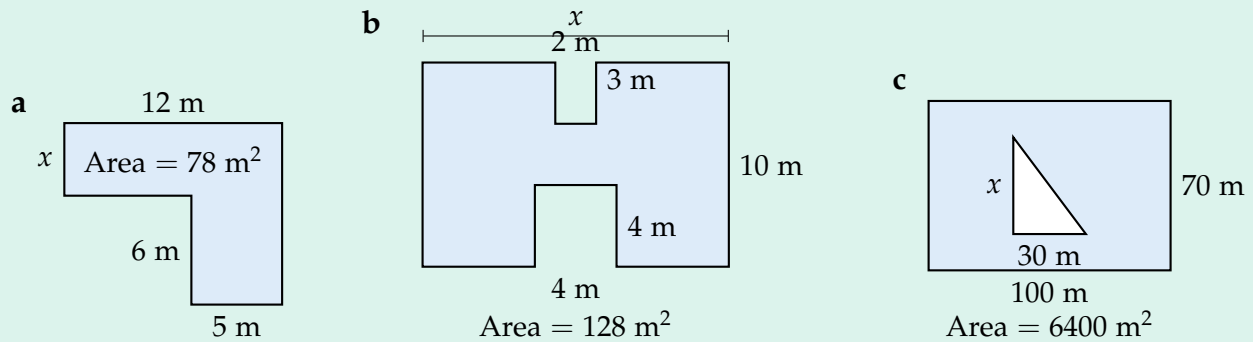
Homework B Missing lengths

Section 4.2

3.

[Maximum mark: 9]

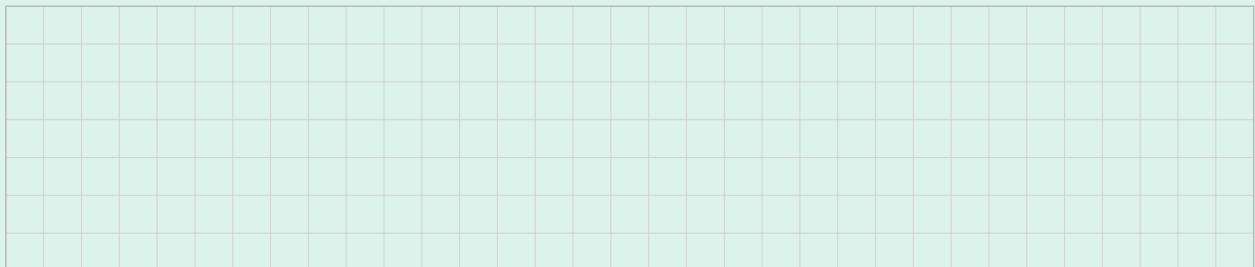
Work out the missing length in each shape.



4.

[Maximum mark: 2]

On the squared paper, draw two **different** compound shapes that each have an area of 36 cm^2 . (Each square is 1 cm by 1 cm – count two small squares as 1 cm.)

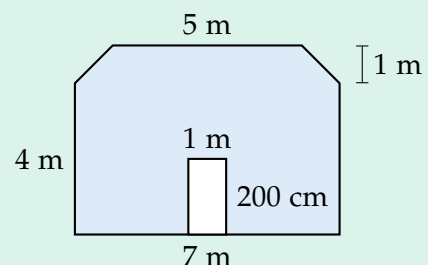


5.

[Maximum mark: 4]

Sam is painting the front of his house (not the door). He has three cans of paint. Each can covers 10 m^2 .

Does he have enough paint? Show your working.



Homework answers B Missing lengths

3. a) Split: top rectangle $12 \times x$ and lower part $5 \times 6 = 30$.

$$12x + 30 = 78 \Rightarrow 12x = 48 \Rightarrow x = \boxed{4 \text{ m}}$$

b) Whole rectangle minus the two notches: $10x - 2 \times 3 - 4 \times 4 = 128$

$$10x - 22 = 128 \Rightarrow 10x = 150 \Rightarrow x = \boxed{15 \text{ m}}$$

c) Rectangle = $100 \times 70 = 7000$. Triangle hole = $7000 - 6400 = 600 \text{ m}^2$.

$$\frac{1}{2} \times 30 \times x = 600 \Rightarrow 15x = 600 \Rightarrow x = \boxed{40 \text{ m}}$$

4. Many answers. Check by counting: e.g. an L-shape made of a 6×4 rectangle and a 3×4 rectangle ($24 + 12 = 36$), or a T-shape made of a 9×2 bar and a 3×6 stem ($18 + 18 = 36$).

5. Rectangle = $7 \times 4 = 28 \text{ m}^2$. Trapezium roof = $\frac{1}{2}(5 + 7) \times 1 = 6 \text{ m}^2$.

Door: $200 \text{ cm} = 2 \text{ m}$, so $1 \times 2 = 2 \text{ m}^2$.

Area to paint = $28 + 6 - 2 = 32 \text{ m}^2$. Three cans cover $3 \times 10 = 30 \text{ m}^2$.

$30 < 32$, so **no** – he does not have enough. He needs $32 \div 10 = 3.2$, i.e. **4 cans**.

Homework B Compound shapes – applied

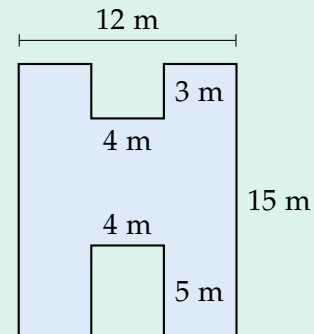
Section 4.2

6.

[Maximum mark: 4]

Mrs Patel is tiling this floor (the H-shape from Q1e). Each tile is a square of side 50 cm.

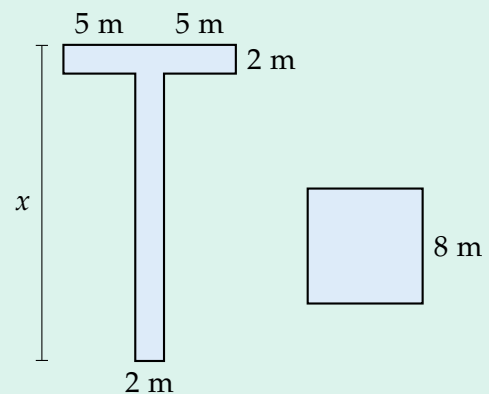
- a) How many tiles does she need?
 b) Each tile costs £3.75. How much will the tiles cost?



[Maximum mark: 4]

7.

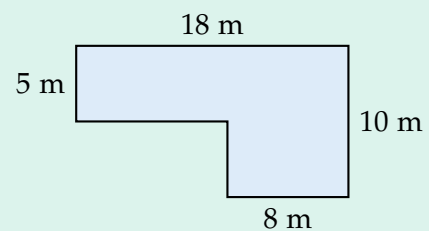
The T-shape and the square have the **same area**. Work out the value of x .



[Maximum mark: 4]

8.

- a) Work out the area of this shape.
 b) Write the area in square centimetres.

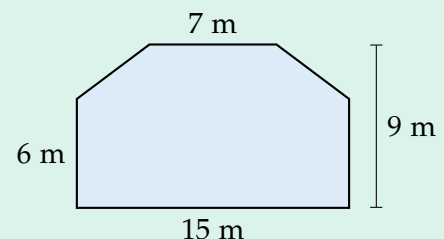


[Maximum mark: 6]

9.

A farmer has a field in this shape. Each sheep needs at least 3 m^2 of grass. A sheep costs about £80.

- a) What is the greatest number of sheep the field can hold?
 b) How much would it cost to buy that many sheep?



Homework answers B Compound shapes – applied

6. a) Area = 148 m^2 (from Q1e). A 50 cm tile covers $0.5 \times 0.5 = 0.25 \text{ m}^2$, so 4 tiles cover 1 m^2 .

Tiles = $148 \times 4 = \boxed{592 \text{ tiles}}$

b) $592 \times £3.75 = \boxed{£2220}$

7. Square = $8 \times 8 = 64 \text{ m}^2$. T-shape: top bar = $(5 + 2 + 5) \times 2 = 24$; stem = $2 \times (x - 2)$.
 $24 + 2(x - 2) = 64 \Rightarrow 2(x - 2) = 40 \Rightarrow x - 2 = 20 \Rightarrow x = \boxed{22 \text{ m}}$

8. a) Top strip $18 \times 5 = 90$; lower part $8 \times (10 - 5) = 40$. Area = $\boxed{130 \text{ m}^2}$

b) $1 \text{ m}^2 = 10\,000 \text{ cm}^2$, so $130 \times 10\,000 = \boxed{1\,300\,000 \text{ cm}^2}$

9. a) Rectangle $15 \times 6 = 90$; trapezium $\frac{1}{2}(7 + 15) \times 3 = 33$. Area = 123 m^2 .

Sheep = $123 \div 3 = \boxed{41 \text{ sheep}}$

b) $41 \times £80 = \boxed{£3280}$

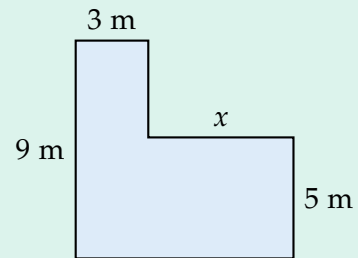
Homework B Compound shapes – exam style Section 4.2

10.

[Maximum mark: 5]

The area of this shape is 57 m^2 .

Work out the perimeter of the shape.

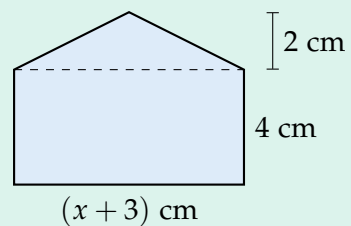


[Maximum mark: 4]

11.

The shape is a rectangle with an isosceles triangle on top. Its area is 60 cm^2 .

Work out the value of x .

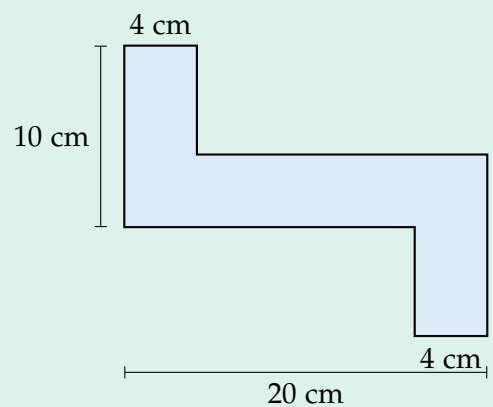


[Maximum mark: 3]

12.

This shape has rotational symmetry of order 2. The arms are 4 cm wide and the middle bar is 4 cm high.

Work out the shaded area.



Homework answers B Compound shapes – exam style

10. Split into the bottom strip, $(3 + x)$ m by 5 m, and the top part, $3 \times (9 - 5) = 12 \text{ m}^2$.

$$5(3 + x) + 12 = 57 \Rightarrow 5(3 + x) = 45 \Rightarrow 3 + x = 9 \Rightarrow x = \boxed{6 \text{ m}}$$

Perimeter = $9 + 5 + 6 + 4 + 3 + 9 = \boxed{36 \text{ m}}$ (bottom = $3 + 6 = 9$ m, inner step = $9 - 5 = 4$ m)

11. Rectangle = $4(x + 3)$; triangle = $\frac{1}{2} \times (x + 3) \times 2 = (x + 3)$.

$$4(x + 3) + (x + 3) = 60 \Rightarrow 5(x + 3) = 60 \Rightarrow x + 3 = 12 \Rightarrow x = \boxed{9}$$

12. Middle bar = $20 \times 4 = 80 \text{ cm}^2$. Each arm sticks out $10 - 4 = 6$ cm beyond the bar:
 $4 \times 6 = 24 \text{ cm}^2$.

$$\text{Total} = 80 + 24 + 24 = \boxed{128 \text{ cm}^2}$$

C Properties of 3D solids

Section 4.3

1.

[Maximum mark: 8]

Copy and complete the table.

Solid	Faces	Edges	Vertices
Cube			
Triangular prism			
Square-based pyramid			
Tetrahedron			
Hexagonal prism			
Pentagonal pyramid			
Cylinder			
Cone			

2.

[Maximum mark: 2]

A solid with flat faces has 7 faces and 10 vertices. Use Euler's rule $F + V - E = 2$ to find the number of edges. Name a solid that fits.

3.

[Maximum mark: 3]

Name each solid.

- a) It has 2 triangular faces and 3 rectangular faces.
- b) It has 1 square face and 4 triangular faces that meet at a point.
- c) It has 1 flat circular face, 1 curved surface and 1 vertex.

4.

[Maximum mark: 3]

An octagon has 8 sides. How many faces, edges and vertices does an **octagonal prism** have?

Answers – C Properties of 3D solids

worked solutions

1.

Solid	Faces	Edges	Vertices
Cube	6	12	8
Triangular prism	5	9	6
Square-based pyramid	5	8	5
Tetrahedron	4	6	4
Hexagonal prism	8	18	12
Pentagonal pyramid	6	10	6
Cylinder	3 (2 flat + 1 curved)	2	0
Cone	2 (1 flat + 1 curved)	1	1

Check with Euler, e.g. hexagonal prism: $8 + 12 - 18 = 2$ ✓

2. $F + V - E = 2 \Rightarrow 7 + 10 - E = 2 \Rightarrow E = \boxed{15 \text{ edges}}$. A **pentagonal prism** fits ($n = 5$: $5 + 2 = 7$ faces, $2 \times 5 = 10$ vertices, $3 \times 5 = 15$ edges).

3. a) Triangular prism b) Square-based pyramid c) Cone

4. Prism with $n = 8$: faces = $8 + 2 = \boxed{10}$, edges = $3 \times 8 = \boxed{24}$, vertices = $2 \times 8 = \boxed{16}$.
Check: $10 + 16 - 24 = 2$ ✓

C Nets of 3D solids

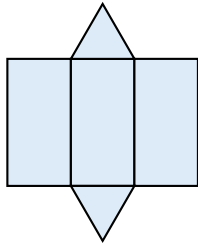
Section 4.3

5.

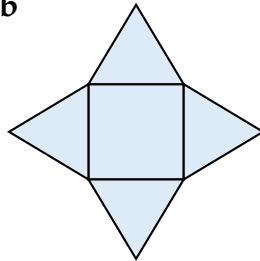
[Maximum mark: 4]

Which 3D solid does each net make?

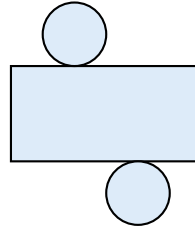
a



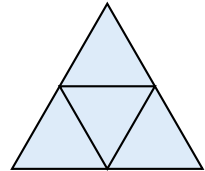
b



c



d

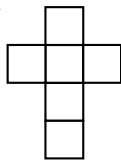


6.

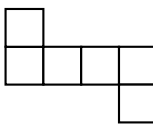
[Maximum mark: 2]

Only two of these are nets of a cube. Which one is **not** a net of a cube? Explain why.

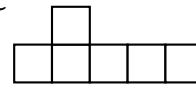
A



B



C

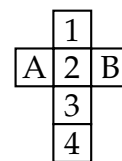


7.

[Maximum mark: 3]

This net is folded to make a cube. Which face will be opposite

a) face 1 b) face A c) face 4?



8.

[Maximum mark: 3]

Sketch a net of a cuboid that measures 4 cm by 3 cm by 2 cm. Label the lengths on your net.

Answers – C Nets of 3D solids

worked solutions

5. a) Triangular prism b) Square-based pyramid c) Cylinder d) Tetrahedron (triangular-based pyramid)

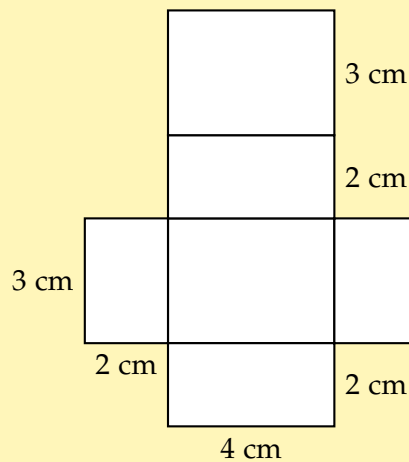
6. C is not a net of a cube. It has **5 squares in a row** – when you fold four of them round into a loop, the fifth square lands on top of the first, so two faces overlap and one face of the cube is left open.

(A and B both fold into a cube.)

7. In the column 1–2–3–4, faces with one square between them are opposite.

a) Face 3 b) Face B c) Face 2

8. One possible net (6 rectangles: two 4×3 , two 4×2 , two 3×2):



Other arrangements are fine as long as there are 6 faces in matching pairs and the edges that join have equal lengths.

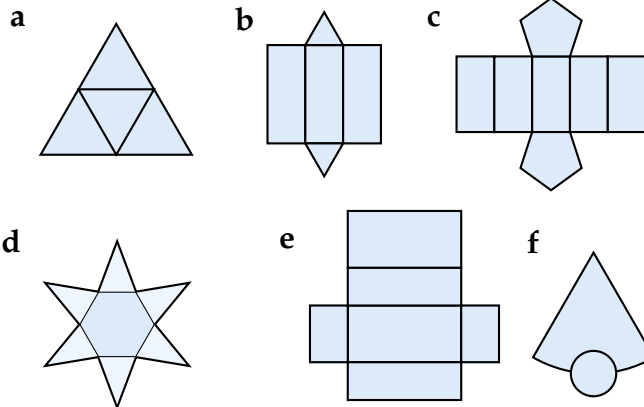
C More nets

Section 4.3

1.

[Maximum mark: 6]

Write down the name of the solid that each net makes.



2.

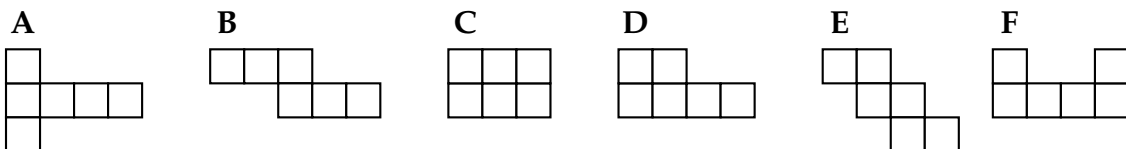
[Maximum mark: 3]

For the solids made from nets **a**, **b** and **c** in Q1, write down the number of faces, edges and vertices. Check each one with $F + V - E = 2$.

3.

[Maximum mark: 3]

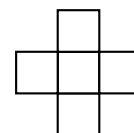
Which of these are nets of a cube? Write down **all** the letters.



4.

[Maximum mark: 1]

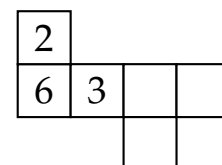
Harley has drawn this shape as the net of a cube. Explain why it is not a correct net.



5.

[Maximum mark: 2]

On a dice, opposite faces add up to 7. Three faces of this dice net are filled in. Write the missing numbers on the other three faces.



Answers – C More nets

worked solutions

1. a) Tetrahedron (triangular-based pyramid) b) Triangular prism c) Pentagonal prism
 d) Hexagonal(-based) pyramid e) Cuboid f) Cone

	Faces	Edges	Vertices	
2. a) Tetrahedron	4	6	4	$(4 + 4 - 6 = 2 \checkmark)$
b) Triangular prism	5	9	6	$(5 + 6 - 9 = 2 \checkmark)$
c) Pentagonal prism	7	15	10	$(7 + 10 - 15 = 2 \checkmark)$

3. **A, B, E**

C is a 2×3 block; D contains a 2×2 block; in F the two top squares would both fold onto the same face (and one face would be missing).

4. It has only **5 squares**, but a cube has **6 faces** – one face is missing.

5. In this net the 1st and 3rd squares of the row are opposite, the 2nd and 4th are opposite, and the top square is opposite the bottom square.

3rd square in the row = $7 - 6 = \boxed{1}$; 4th square = $7 - 3 = \boxed{4}$; bottom square = $7 - 2 = \boxed{5}$.

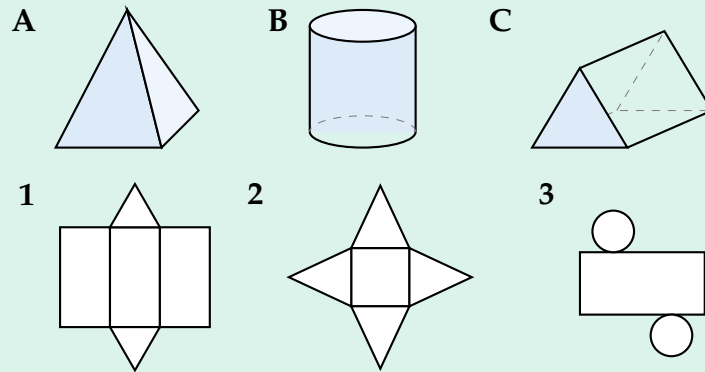
Homework C 3D solids and nets

Section 4.3

1.

[Maximum mark: 6]

The diagram shows three 3D solids, **A**, **B** and **C**, and three nets, **1**, **2** and **3**.



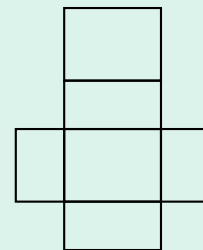
- a) Match each solid to its net. b) Write down the name of solid C.
 c) How many faces does solid A have? d) How many vertices does solid A have?
 e) How many edges does solid C have?

2.

[Maximum mark: 5]

The diagram shows the net of a solid.

- a) Write down the name of the solid.
 b) The net has one line of symmetry. Describe where it is.
 c) How many faces, d) vertices and e) edges does the solid have?



3.

[Maximum mark: 1]

Here is the net of a 3D shape. Which shape is it?

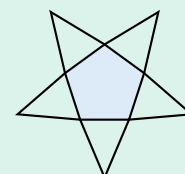
Cylinder Pyramid Cube Cone



4.

[Maximum mark: 1]

Which 3D shape does this net make?



Homework answers C 3D solids and nets

1. a) A (square-based pyramid) – net 2; B (cylinder) – net 3; C (triangular prism) – net 1.
b) Triangular prism c) 5 faces (1 square + 4 triangles) d) 5 vertices (4 at the base + 1 at the top) e) 9 edges
2. a) Cuboid b) A vertical line down the middle of the long column of rectangles.
c) 6 faces d) 8 vertices e) 12 edges
3. **Cone** – a curved sector (it rolls into the curved surface) and one circle for the base.
4. A **pentagonal(-based) pyramid**: 1 pentagon base and 5 triangles that meet at the top.

Homework C Drawing nets

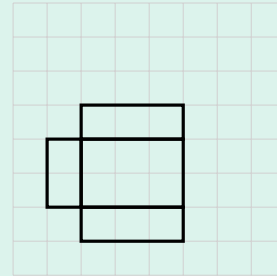
Section 4.3

5.

[Maximum mark: 1]

Part of the net of a cuboid is drawn on the centimetre grid. The cuboid is 3 cm by 2 cm by 1 cm.

Complete the net.



6.

[Maximum mark: 2]

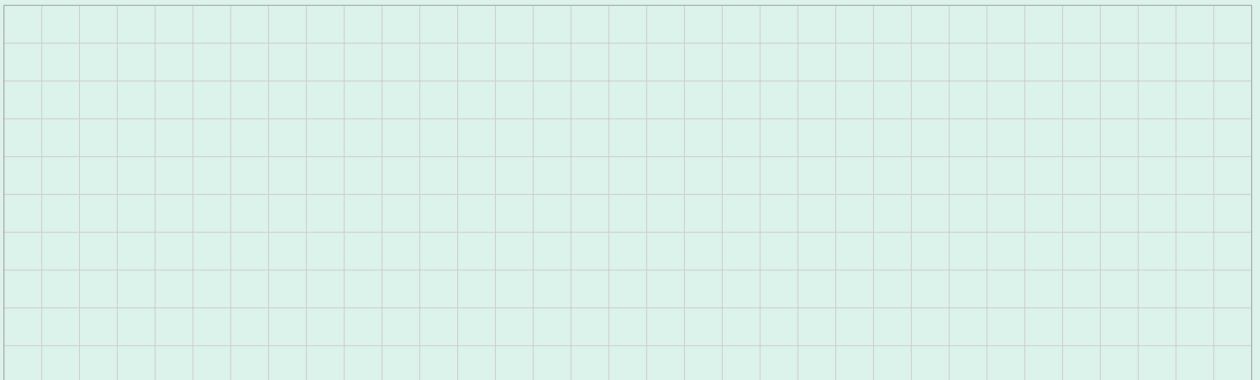
Sketch a net of a square-based pyramid.



7.

[Maximum mark: 2]

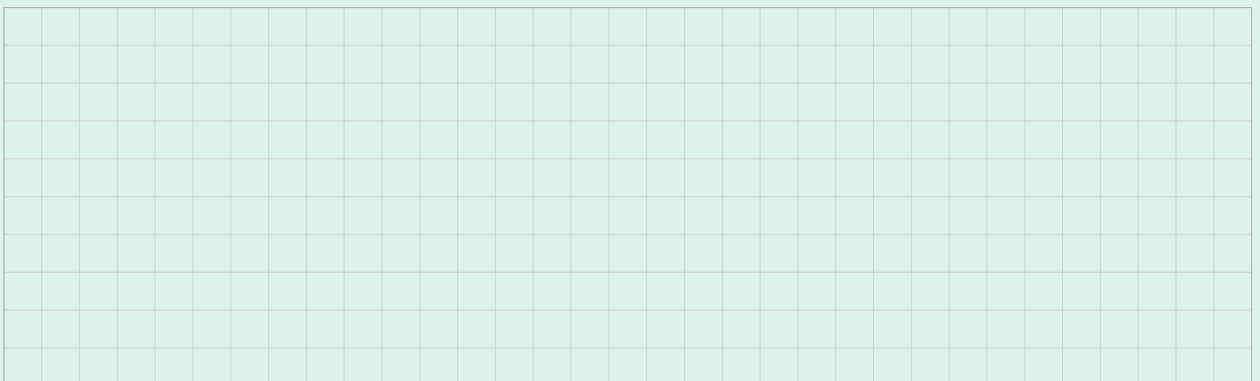
A cube has edges of 3 cm. On the grid (each small square is 0.5 cm), draw an accurate net of the cube.



8.

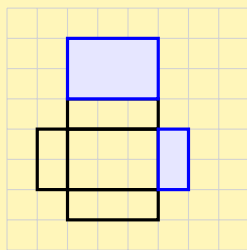
[Maximum mark: 3]

A cuboid measures 4 cm by 3 cm by 1 cm. On the grid, draw an accurate net of the cuboid.



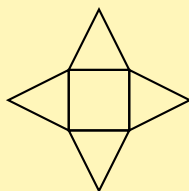
Homework answers C Drawing nets

5. Add the **top** (3×2) and the **right-hand end** (1×2) – shown in blue:

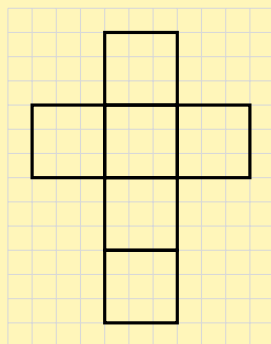


(The top could instead go below the front face, and the end could go on the right of the top – any arrangement that folds correctly is fine.)

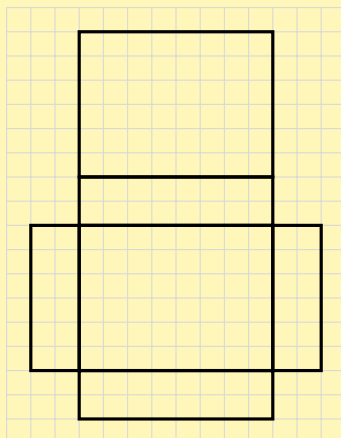
6. A square with a triangle on each of its four sides, for example:



7. Six 3 cm by 3 cm squares (6 small grid squares across each) arranged as a cube net, e.g. a column of four with one square on each side of the second (drawn smaller here):



8. Six faces: two 4×3 , two 4×1 , two 3×1 (8, 6 and 2 small squares long), e.g.:



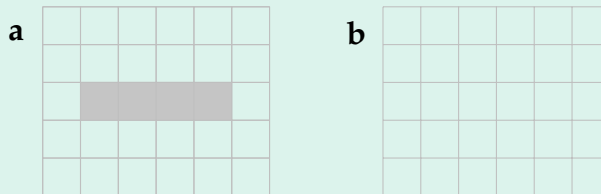
Homework C Nets – which is which?

Section 4.3

9.

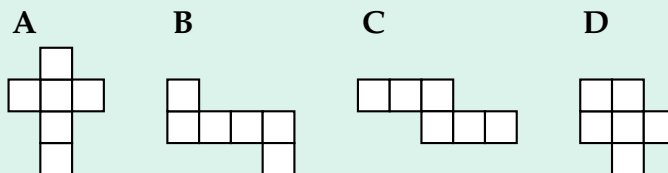
[Maximum mark: 2]

- a) Shade two more squares so that the shaded shape is a net of a cube.
 b) On the second grid, shade six squares to make a **different** net of a cube.



10.

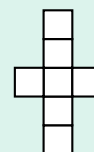
[Maximum mark: 1]

Which of these is **not** a net of a cube?

11.

[Maximum mark: 1]

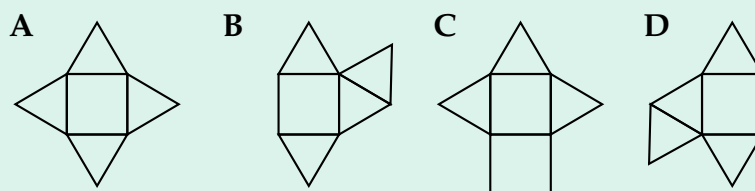
Harley says this is the net of a cube. Explain why he is wrong.



12.

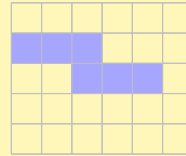
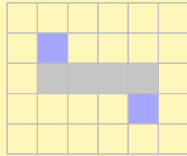
[Maximum mark: 1]

Three of these are nets of a square-based pyramid. Which one is **not**? Explain why.



Homework answers C Nets – which is which?

9. a) Shade one square **above** and one square **below** the row of four (they can be above/below any of the four squares), e.g. above the first and below the last.
 b) Any other cube net, e.g. the “staircase” 3–3 net: three in a row, then three in a row shifted along by two.



10. **D** – it contains a 2×2 block of squares. When folded, two faces would overlap.
 11. The net has **7 squares** but a cube has only **6 faces**. (Also, five squares in a line can't fold – the fifth would overlap the first.)
 12. **C** – it has **two squares** and only three triangles. A square-based pyramid has 1 square face and 4 triangular faces.
 (B and D work: the fourth triangle is attached to the sloping side of another triangle and folds round into place.)

Homework C Nets – problem solving

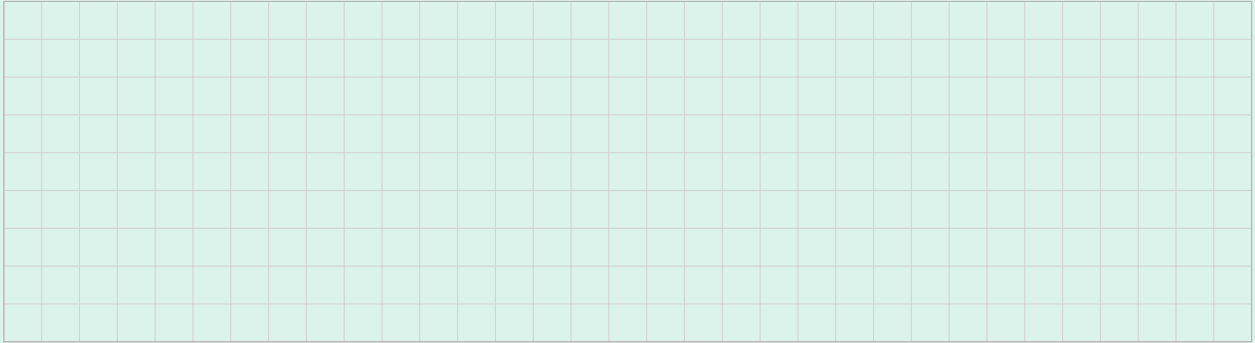
Section 4.3

13.

[Maximum mark: 5]

A cuboid has length 5 cm, width 3 cm and height 1 cm.

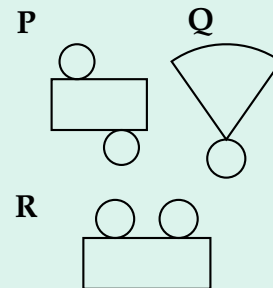
- a) On the grid, draw an accurate net of the cuboid. b) Work out the total surface area of the cuboid.



14.

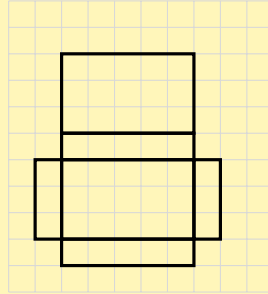
[Maximum mark: 3]

- a) Which solid is made from net **P**?
 b) Which solid is made from net **Q**?
 c) Kai drew shape **R** as the net of a cylinder. Explain why it is not a correct net.



Homework answers C Nets – problem solving

13. a) Faces: two 5×3 , two 5×1 , two 3×1 , e.g. (1 square = 1 cm):



b) $SA = 2(5 \times 3 + 5 \times 1 + 3 \times 1) = 2(15 + 5 + 3) = \boxed{46 \text{ cm}^2}$

14. a) Cylinder b) Cone

c) Both circles are on the **same side** of the rectangle. When the rectangle is rolled into a tube, both circles would cover the same end – the other end would be open. One circle must go on the top edge and one on the bottom edge.

Homework C Nets – puzzles

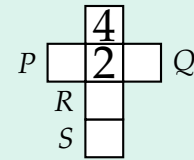
Section 4.3

15.

[Maximum mark: 2]

On a dice, the numbers on opposite faces add up to 7.

This is a net of a dice. Two faces are filled in. Write the numbers 1 to 6 that go on the other four faces.

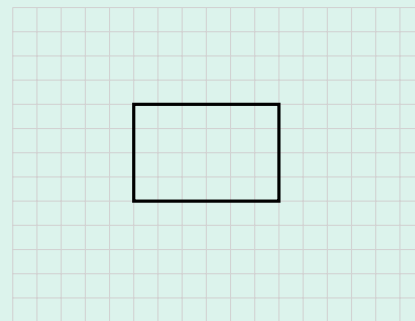


16.

[Maximum mark: 3]

A triangular prism has a right-angled triangle cross-section with sides 3 cm, 4 cm and 5 cm. It is 6 cm long.

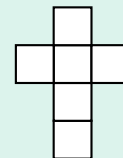
One rectangular face (4 cm by 6 cm) is drawn on the centimetre grid. Complete the net.



17.

[Maximum mark: 3]

Write 6 **different** whole numbers on the faces of this cube net so that the numbers on **opposite** faces of the cube multiply to give 60.

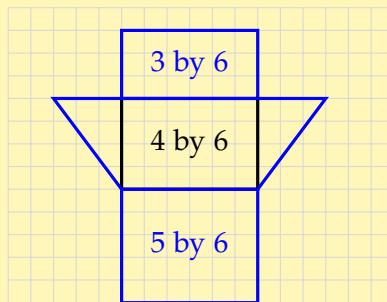


Homework answers C Nets – puzzles

15. Opposite faces in this net: top & third square down, second & bottom square, P & Q .

Face opposite 4 (R) = $7 - 4 = \boxed{3}$; face opposite 2 (S) = $7 - 2 = \boxed{5}$; P and Q are $\boxed{1}$ and $\boxed{6}$ (either way round).

16. A strip of three rectangles, 3×6 , 4×6 and 5×6 , with the two right-angled triangles (legs 3 cm and 4 cm) on the ends of the strip, e.g.:



The 3 cm, 4 cm and 5 cm rectangles join along their 6 cm edges; each triangle is attached to a short edge.

17. Pick three pairs that multiply to 60, e.g. 1×60 , 2×30 , 3×20 (or 4×15 , 5×12 , 6×10).

Put each pair on **opposite** faces: top square & third square down; second square & bottom square; left & right squares.

e.g. top = 1, third = 60; second = 2, bottom = 30; left = 3, right = 20.

Homework C More nets – drawing

Section 4.3

18.

[Maximum mark: 2]

Sketch a net of a triangular prism. Label which faces are triangles and which are rectangles.



19.

[Maximum mark: 3]

A cuboid measures 5 cm by 2 cm by 2 cm. Draw an accurate net of the cuboid on the grid (each small square is 0.5 cm).

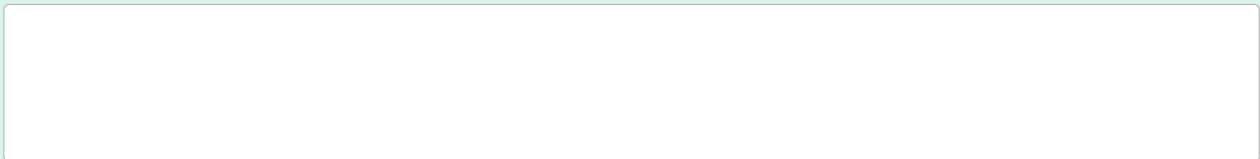


20.

[Maximum mark: 4]

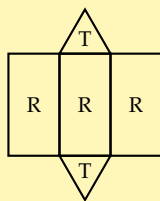
An open box (no lid) is a cuboid 6 cm long, 4 cm wide and 3 cm high.

a) Sketch a net of the box. **b)** Work out the area of card needed.



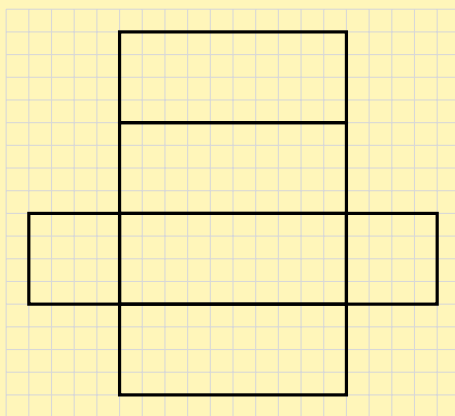
Homework answers C More nets – drawing

18. Three rectangles in a strip with a triangle on each end of the middle rectangle, e.g.:



2 triangles (T) and 3 rectangles (R) – other arrangements are fine if they fold correctly.

19. The faces are four 5×2 rectangles and two 2×2 squares. On the grid these are 10 by 4 small squares and 4 by 4 small squares, e.g.:



20. a) Base 6×4 with the four sides folded up: two 6×3 and two 4×3 (no top).

b) $6 \times 4 + 2 \times 6 \times 3 + 2 \times 4 \times 3 = 24 + 36 + 24 = \boxed{84 \text{ cm}^2}$

Homework C More nets – problems

Section 4.3

21.

[Maximum mark: 3]

A square-based pyramid has a base of side 6 cm. Each triangular face has a perpendicular height of 5 cm.

- a)** Sketch its net. **b)** Work out the total area of the net.

22.

[Maximum mark: 3]

The net of a cube has a total area of 150 cm^2 . Work out **a)** the length of one edge **b)** the volume of the cube.

23.

[Maximum mark: 3]

The net of a triangular prism is made from two equilateral triangles of side 4 cm and three rectangles 4 cm by 10 cm.

When it is folded, work out **a)** the number of edges **b)** the total length of all the edges.

24.

[Maximum mark: 4]

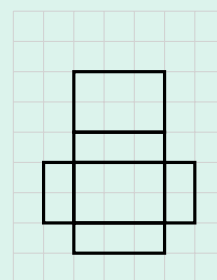
- a)** A net is made from 2 hexagons and 6 rectangles. Name the solid.
b) A net is made from 1 octagon and 8 triangles. Name the solid.
c) Write down the number of faces, edges and vertices of each solid.

25.

[Maximum mark: 3]

This net is drawn on a centimetre grid. It folds to make a cuboid.

Work out **a)** the volume **b)** the surface area of the cuboid.



Homework answers C More nets – problems

21. a) A 6 cm square with a triangle (base 6 cm, height 5 cm) on each side.

b) $6 \times 6 + 4 \times \frac{1}{2} \times 6 \times 5 = 36 + 60 = \boxed{96 \text{ cm}^2}$

22. a) One face = $150 \div 6 = 25 \text{ cm}^2$, edge = $\boxed{5 \text{ cm}}$ **b)** $5^3 = \boxed{125 \text{ cm}^3}$

23. a) $\boxed{9}$ edges **b)** 6 edges of 4 cm (around the triangles) and 3 edges of 10 cm:
 $24 + 30 = \boxed{54 \text{ cm}}$

24. a) Hexagonal prism **b)** Octagonal(-based) pyramid

c) Hexagonal prism: 8 faces, 18 edges, 12 vertices. Octagonal pyramid: 9 faces, 16 edges, 9 vertices.

25. The cuboid is 3 cm by 2 cm by 1 cm.

a) $3 \times 2 \times 1 = \boxed{6 \text{ cm}^3}$ **b)** $2(3 \times 2 + 3 \times 1 + 2 \times 1) = 2 \times 11 = \boxed{22 \text{ cm}^2}$ (count the squares: $6 + 3 + 6 + 3 + 2 + 2 = 22 \checkmark$)

D Surface area

Section 4.4

1.

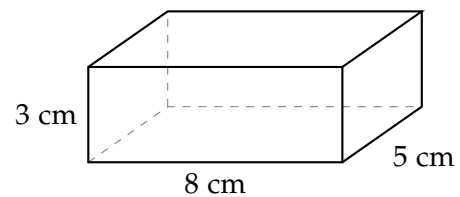
[Maximum mark: 2]

Work out the surface area of a cube with side length 4 cm.

2.

[Maximum mark: 3]

Work out the surface area of this cuboid.

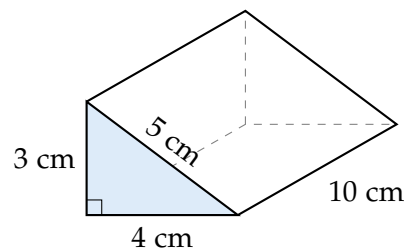


3.

[Maximum mark: 4]

The cross-section of this prism is a right-angled triangle with sides 3 cm, 4 cm and 5 cm. The prism is 10 cm long.

Work out its surface area.



4.

[Maximum mark: 3]

A cube has a total surface area of 150 cm^2 . Work out the length of one edge of the cube.

Answers – D Surface area

worked solutions

1. One face = $4 \times 4 = 16 \text{ cm}^2$. $SA = 6 \times 16 = \boxed{96 \text{ cm}^2}$

2. Front = $8 \times 3 = 24$, top = $8 \times 5 = 40$, side = $5 \times 3 = 15 \text{ cm}^2$.

$$SA = 2(24 + 40 + 15) = 2 \times 79 = \boxed{158 \text{ cm}^2}$$

3. Two triangles: $2 \times \frac{1}{2} \times 4 \times 3 = 12 \text{ cm}^2$.

Three rectangles: $(3 + 4 + 5) \times 10 = 120 \text{ cm}^2$ (perimeter of cross-section $\times$ length).

$$SA = 12 + 120 = \boxed{132 \text{ cm}^2}$$

4. One face = $150 \div 6 = 25 \text{ cm}^2$. Edge = $\sqrt{25} = \boxed{5 \text{ cm}}$

D Surface area – part 2

Section 4.4

5.

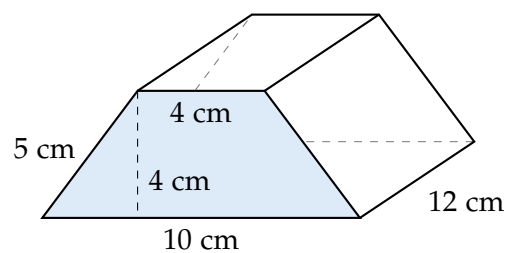
[Maximum mark: 3]

An open box (no lid) is a cuboid 30 cm long, 20 cm wide and 10 cm high.
Work out the area of card needed to make the box.

6.

[Maximum mark: 5]

The cross-section of this prism is an isosceles trapezium. The parallel sides are 4 cm and 10 cm, the height is 4 cm and the sloping sides are 5 cm. The prism is 12 cm long.
Work out the surface area.



7.

[Maximum mark: 4]

A storage box is a cuboid 1.2 m by 0.8 m by 0.5 m. Every face is painted. One tin of paint covers 2 m^2 .
How many tins are needed?

8.

[Maximum mark: 3]

A cuboid has a surface area of 94 cm^2 . Its length is 5 cm and its width is 4 cm. Work out its height.

Answers – D Surface area – part 2

worked solutions

5. Base = $30 \times 20 = 600$. Front and back = $2 \times 30 \times 10 = 600$. Two ends = $2 \times 20 \times 10 = 400$.

No lid, so total = $600 + 600 + 400 = \boxed{1600 \text{ cm}^2}$

6. Two trapezium ends: $2 \times \frac{1}{2}(4 + 10) \times 4 = 2 \times 28 = 56 \text{ cm}^2$.

Perimeter of cross-section = $4 + 10 + 5 + 5 = 24 \text{ cm}$, rectangles = $24 \times 12 = 288 \text{ cm}^2$.

SA = $56 + 288 = \boxed{344 \text{ cm}^2}$

7. $SA = 2(1.2 \times 0.8 + 1.2 \times 0.5 + 0.8 \times 0.5) = 2(0.96 + 0.6 + 0.4) = 2 \times 1.96 = 3.92 \text{ m}^2$.

Tins: $3.92 \div 2 = 1.96$, so $\boxed{2 \text{ tins}}$.

8. $2(5 \times 4 + 5h + 4h) = 94 \Rightarrow 20 + 9h = 47 \Rightarrow 9h = 27 \Rightarrow h = \boxed{3 \text{ cm}}$

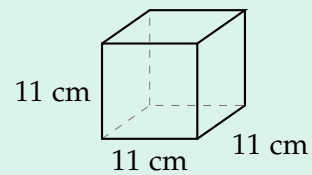
Homework D Surface area

Section 4.4

1.

[Maximum mark: 2]

Calculate the surface area of this cube.



2.

[Maximum mark: 2]

The total surface area of a cube is 337.5 cm^2 . Work out the length of each edge.

3.

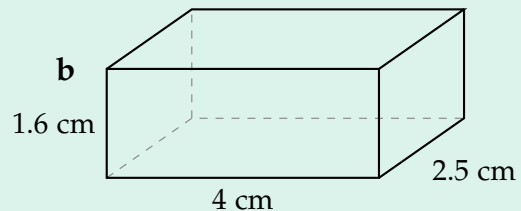
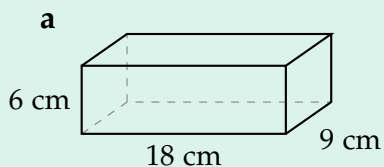
[Maximum mark: 2]

A cube has edges of 6.5 cm. What is its surface area?

4.

[Maximum mark: 6]

Work out the surface area of each cuboid.

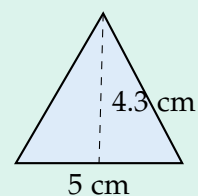


5.

[Maximum mark: 3]

All four faces of this pyramid are the same equilateral triangle, with base 5 cm and perpendicular height 4.3 cm (including the base).

Work out its surface area.



Homework answers D Surface area

1. One face = $11 \times 11 = 121 \text{ cm}^2$. $SA = 6 \times 121 = \boxed{726 \text{ cm}^2}$
2. One face = $337.5 \div 6 = 56.25 \text{ cm}^2$. Edge = $\sqrt{56.25} = \boxed{7.5 \text{ cm}}$
3. $6 \times 6.5 \times 6.5 = 6 \times 42.25 = \boxed{253.5 \text{ cm}^2}$
4. a) $2(18 \times 6 + 18 \times 9 + 9 \times 6) = 2(108 + 162 + 54) = 2 \times 324 = \boxed{648 \text{ cm}^2}$
b) $2(4 \times 1.6 + 4 \times 2.5 + 2.5 \times 1.6) = 2(6.4 + 10 + 4) = 2 \times 20.4 = \boxed{40.8 \text{ cm}^2}$
5. One face = $\frac{1}{2} \times 5 \times 4.3 = 10.75 \text{ cm}^2$. 4 faces: $4 \times 10.75 = \boxed{43 \text{ cm}^2}$

Homework D Pyramids & composite solids

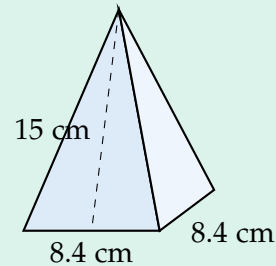
Section 4.4

6.

[Maximum mark: 3]

This square-based pyramid has a base of side 8.4 cm. Each triangular face has a perpendicular height of 15 cm (shown dashed).

Work out the total surface area.

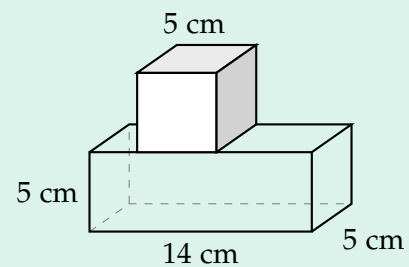


7.

[Maximum mark: 4]

A cube of side 5 cm is glued on top of a cuboid 14 cm by 5 cm by 5 cm. The front of the cube lines up with the front of the cuboid.

Work out the surface area of the solid.

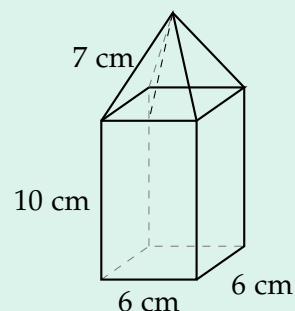


8.

[Maximum mark: 4]

A square-based pyramid sits on top of a cuboid. The cuboid is 6 cm by 6 cm by 10 cm tall. Each triangular face of the pyramid has a perpendicular height of 7 cm.

Work out the total surface area.

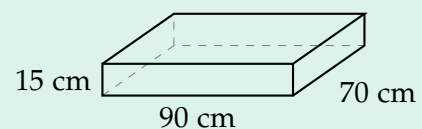


9.

[Maximum mark: 4]

A sofa has 5 identical cushions. Each is a cuboid 90 cm by 70 cm by 15 cm. Every face of every cushion is sprayed. One can of spray covers 3 m².

How many cans are needed?



Homework answers D Pyramids & composite solids

6. Base = $8.4 \times 8.4 = 70.56 \text{ cm}^2$. Each triangle = $\frac{1}{2} \times 8.4 \times 15 = 63 \text{ cm}^2$, four of them = 252 cm^2 .

$$SA = 70.56 + 252 = \boxed{322.56 \text{ cm}^2}$$

7. Cuboid: front & back $2 \times 14 \times 5 = 140$; ends $2 \times 5 \times 5 = 50$; bottom $14 \times 5 = 70$; top (part not covered) $70 - 25 = 45$. Cuboid total = 305 cm^2 .

Cube: 5 faces showing (not the bottom) = $5 \times 25 = 125 \text{ cm}^2$.

$$\text{Total} = 305 + 125 = \boxed{430 \text{ cm}^2}$$

8. Cuboid without its top: base $6 \times 6 = 36$ + four sides $4 \times 6 \times 10 = 240 \Rightarrow 276 \text{ cm}^2$.

Pyramid without its base: $4 \times \frac{1}{2} \times 6 \times 7 = 84 \text{ cm}^2$.

$$\text{Total} = 276 + 84 = \boxed{360 \text{ cm}^2}$$

9. One cushion: $2(90 \times 70 + 90 \times 15 + 70 \times 15) = 2(6300 + 1350 + 1050) = 17\,400 \text{ cm}^2$.

5 cushions: $87\,000 \text{ cm}^2 = 87\,000 \div 10\,000 = 8.7 \text{ m}^2$.

Cans: $8.7 \div 3 = 2.9$, so $\boxed{3 \text{ cans}}$.

Homework D Surface area of prisms

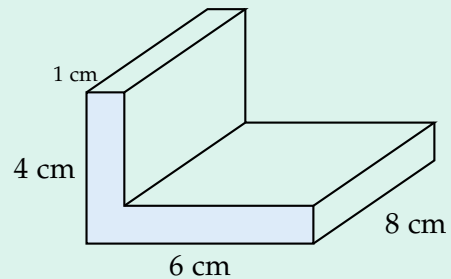
Section 4.4

10.

[Maximum mark: 4]

This prism has an L-shaped cross-section (all corners are right angles). The arms are 1 cm thick. The prism is 8 cm long.

Work out the total surface area.

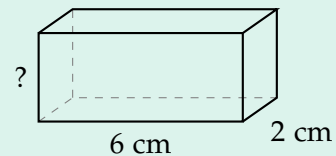


11.

[Maximum mark: 3]

This solid cuboid has a base 6 cm by 2 cm. Its volume is 60 cm^3 .

Work out its total surface area.



12.

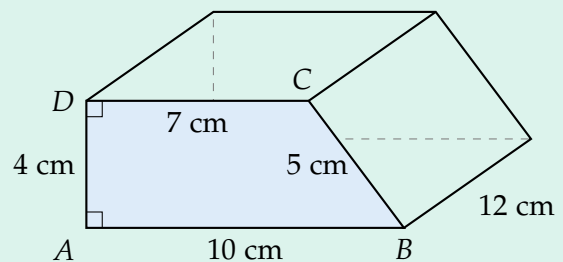
[Maximum mark: 5]

The cross-section of this solid prism is the trapezium $ABCD$.

Angle $DAB = \text{angle } ADC = 90^\circ$.

$AB = 10 \text{ cm}$, $DC = 7 \text{ cm}$, $AD = 4 \text{ cm}$ and $BC = 5 \text{ cm}$. The prism is 12 cm long.

Work out the total surface area.



Homework answers D Surface area of prisms

10. Cross-section: $6 \times 1 + 1 \times 3 = 9 \text{ cm}^2$ (two of them = 18 cm^2).

Perimeter of cross-section = $6 + 1 + 5 + 3 + 1 + 4 = 20 \text{ cm}$. Rectangles = $20 \times 8 = 160 \text{ cm}^2$.

$$SA = 18 + 160 = \boxed{178 \text{ cm}^2}$$

11. Height = $60 \div (6 \times 2) = 5 \text{ cm}$.

$$SA = 2(6 \times 2 + 6 \times 5 + 2 \times 5) = 2(12 + 30 + 10) = \boxed{104 \text{ cm}^2}$$

12. Two trapezium ends: $2 \times \frac{1}{2}(10 + 7) \times 4 = 68 \text{ cm}^2$.

Perimeter = $10 + 5 + 7 + 4 = 26 \text{ cm}$, so the four rectangles = $26 \times 12 = 312 \text{ cm}^2$.

$$SA = 68 + 312 = \boxed{380 \text{ cm}^2}$$

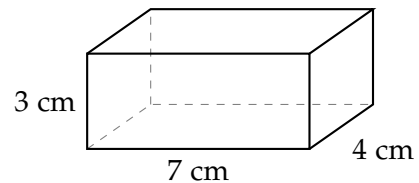
E Volume of cuboids and prisms

Section 4.5

1.

[Maximum mark: 2]

Work out the volume of this cuboid. Include units.



2.

[Maximum mark: 2]

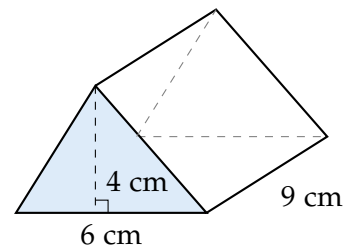
Work out the volume of a cube with edges of 6 cm.

3.

[Maximum mark: 3]

The cross-section of this prism is a triangle with base 6 cm and perpendicular height 4 cm. The prism is 9 cm long.

Work out the volume.



4.

[Maximum mark: 2]

A prism has a cross-sectional area of 21 cm^2 and a length of 6 cm. Find its volume.

5.

[Maximum mark: 2]

A prism has a cross-sectional area of 55 cm^2 and a volume of 330 cm^3 . Find the length of the prism.

Answers – E Volume

worked solutions

1. $V = 7 \times 4 \times 3 = \boxed{84 \text{ cm}^3}$

2. $V = 6^3 = 6 \times 6 \times 6 = \boxed{216 \text{ cm}^3}$

3. Cross-section $= \frac{1}{2} \times 6 \times 4 = 12 \text{ cm}^2$. $V = 12 \times 9 = \boxed{108 \text{ cm}^3}$

4. $V = 21 \times 6 = \boxed{126 \text{ cm}^3}$

5. Length $= 330 \div 55 = \boxed{6 \text{ cm}}$

E Volume – part 2

Section 4.5

6.

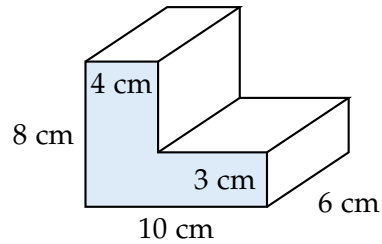
[Maximum mark: 2]

A cube has a volume of 64 cm^3 . How long is each edge?

7.

[Maximum mark: 4]

This prism has an L-shaped cross-section (dimensions shown) and is 6 cm long. Work out its volume.



8.

[Maximum mark: 3]

A prism has a trapezium cross-section with parallel sides 5 cm and 9 cm and height 4 cm. The prism is 10 cm long. Work out its volume.

9.

[Maximum mark: 2]

A cuboid has volume 240 cm^3 . Its length is 10 cm and its width is 6 cm. Find its height.

10.

[Maximum mark: 3]

A fish tank is a cuboid 60 cm long, 30 cm wide and 40 cm tall. It is filled with water to 5 cm below the top. How many litres of water are in the tank? (1 litre = 1000 cm^3)

11.

[Maximum mark: 2]

How many 2 cm cubes will fit exactly into a box measuring 10 cm by 8 cm by 6 cm?

Answers – E Volume – part 2

worked solutions

6. $4 \times 4 \times 4 = 64$, so each edge = $\sqrt[3]{64} = \boxed{4 \text{ cm}}$

7. Cross-section (split into rectangles) = $10 \times 3 + 4 \times 5 = 30 + 20 = 50 \text{ cm}^2$.

$V = 50 \times 6 = \boxed{300 \text{ cm}^3}$

8. Cross-section = $\frac{1}{2}(5 + 9) \times 4 = 28 \text{ cm}^2$. $V = 28 \times 10 = \boxed{280 \text{ cm}^3}$

9. $10 \times 6 \times h = 240 \Rightarrow 60h = 240 \Rightarrow h = \boxed{4 \text{ cm}}$

10. Water depth = $40 - 5 = 35 \text{ cm}$. $V = 60 \times 30 \times 35 = 63\,000 \text{ cm}^3$.

$63\,000 \div 1000 = \boxed{63 \text{ litres}}$

11. Along: $10 \div 2 = 5$, across: $8 \div 2 = 4$, up: $6 \div 2 = 3$. $5 \times 4 \times 3 = \boxed{60 \text{ cubes}}$

(Check: $480 \div 8 = 60 \checkmark$)

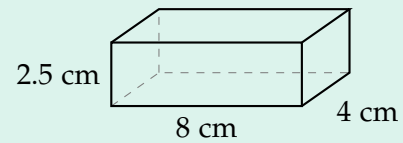
Homework E Volume of cuboids and prisms

Section 4.5

1.

[Maximum mark: 2]

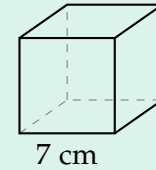
Find the volume of this cuboid. Include units.



2.

[Maximum mark: 2]

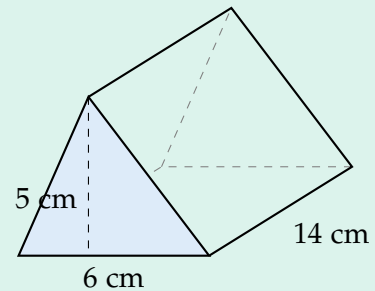
Find the volume of this cube. Include units.



3.

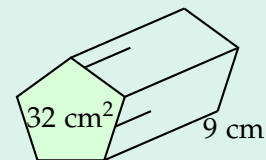
[Maximum mark: 3]

Find the volume of this triangular prism.



4.

[Maximum mark: 2]

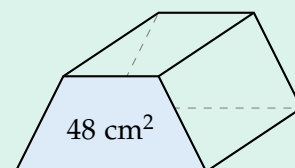
This prism has a cross-sectional area of 32 cm^2 and a length of 9 cm. Find its volume.


5.

[Maximum mark: 2]

This trapezium prism has a cross-sectional area of 48 cm^2 . Its volume is 576 cm^3 .

Find the length of the prism.



Homework answers E Volume

1. $V = 8 \times 4 \times 2.5 = \boxed{80 \text{ cm}^3}$

2. $V = 7^3 = 7 \times 7 \times 7 = \boxed{343 \text{ cm}^3}$

3. Cross-section $= \frac{1}{2} \times 6 \times 5 = 15 \text{ cm}^2$. $V = 15 \times 14 = \boxed{210 \text{ cm}^3}$

4. $V = 32 \times 9 = \boxed{288 \text{ cm}^3}$

5. Length $= 576 \div 48 = \boxed{12 \text{ cm}}$

Homework E Volume of prisms

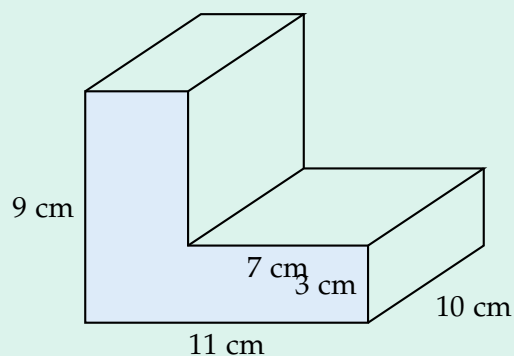
Section 4.5

6.

[Maximum mark: 4]

This prism has a step-shaped cross-section. All corners are right angles. The prism is 10 cm long.

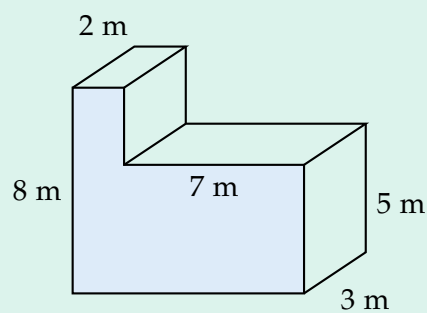
Work out its volume.



7.

[Maximum mark: 3]

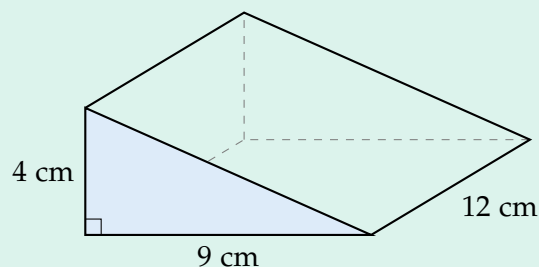
Work out the volume of this prism. All corners are right angles.



8.

[Maximum mark: 3]

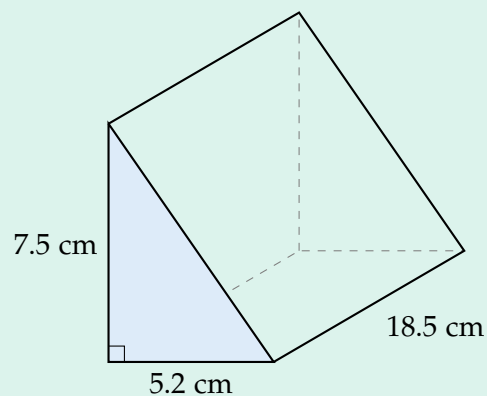
Find the volume of this triangular prism.



9.

[Maximum mark: 3]

Find the volume of this triangular prism.



Homework answers E Volume of prisms

6. Cross-section: bottom strip $11 \times 3 = 33$ + tall part $4 \times (9 - 3) = 24 \Rightarrow 57 \text{ cm}^2$.

$$V = 57 \times 10 = \boxed{570 \text{ cm}^3}$$

7. Cross-section: bottom $9 \times 5 = 45$ (width = $2 + 7 = 9 \text{ m}$) + top $2 \times (8 - 5) = 6 \Rightarrow 51 \text{ m}^2$.

$$V = 51 \times 3 = \boxed{153 \text{ m}^3}$$

8. Cross-section = $\frac{1}{2} \times 9 \times 4 = 18 \text{ cm}^2$. $V = 18 \times 12 = \boxed{216 \text{ cm}^3}$

9. Cross-section = $\frac{1}{2} \times 5.2 \times 7.5 = 19.5 \text{ cm}^2$. $V = 19.5 \times 18.5 = \boxed{360.75 \text{ cm}^3}$

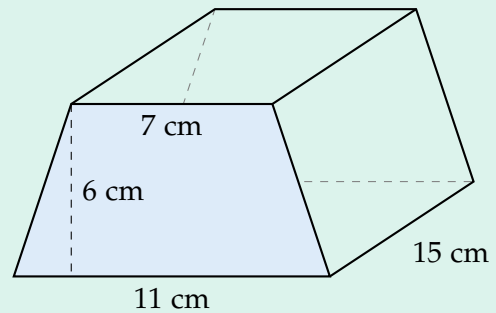
Homework E Volume – harder prisms

Section 4.5

10.

[Maximum mark: 4]

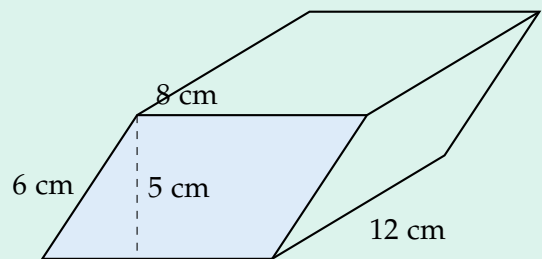
Find the volume of this trapezium prism.



11.

[Maximum mark: 3]

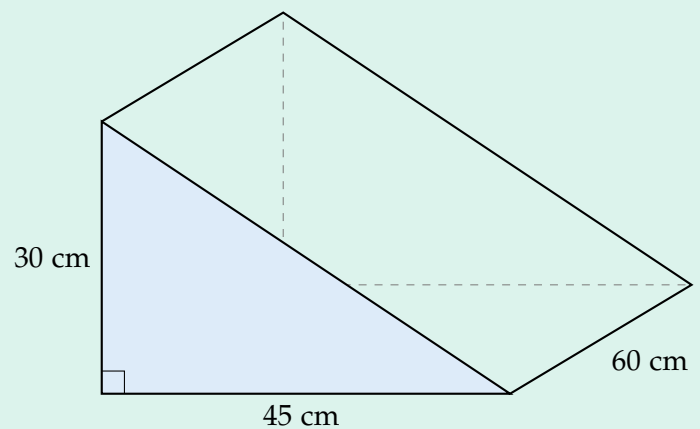
The cross-section of this prism is a parallelogram. Find the volume.



12.

[Maximum mark: 4]

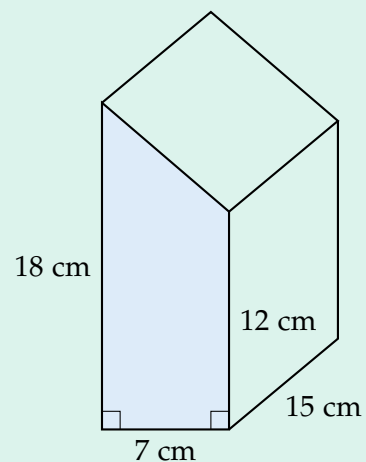
This solid metal prism has a right-angled triangle cross-section. It is melted down and made into solid cubes with edges of 4 cm. How many cubes can be made?



13.

[Maximum mark: 3]

The front face of this prism is a trapezium with two vertical parallel sides of 18 cm and 12 cm, 7 cm apart. The prism is 15 cm deep. Work out the volume.



Homework answers E Volume – harder prisms

10. Cross-section = $\frac{1}{2}(7 + 11) \times 6 = 54 \text{ cm}^2$. $V = 54 \times 15 = \boxed{810 \text{ cm}^3}$

11. Cross-section = base $\times$ perpendicular height = $8 \times 5 = 40 \text{ cm}^2$ (not 8×6).
 $V = 40 \times 12 = \boxed{480 \text{ cm}^3}$

12. Prism: $\frac{1}{2} \times 45 \times 30 \times 60 = 675 \times 60 = 40\,500 \text{ cm}^3$. One cube = $4^3 = 64 \text{ cm}^3$.
 $40\,500 \div 64 = 632.8\dots$, so $\boxed{632 \text{ cubes}}$ (round **down** – you can't make a part cube).

13. Cross-section (trapezium) = $\frac{1}{2}(18 + 12) \times 7 = 105 \text{ cm}^2$. $V = 105 \times 15 = \boxed{1575 \text{ cm}^3}$

Homework E Volume – problem solving

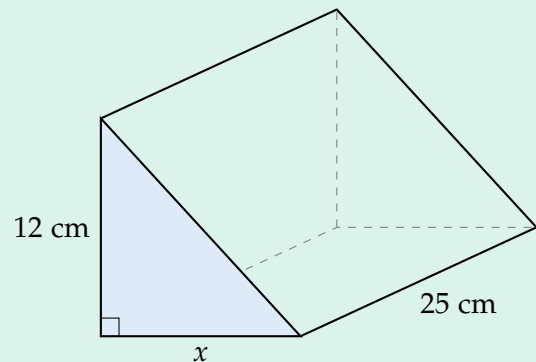
Section 4.5

14.

[Maximum mark: 3]

The triangle at the end of this prism has perpendicular height 12 cm and base x cm. The prism is 25 cm long and its volume is 1650 cm^3 .

Work out x .



15.

[Maximum mark: 3]

A carton is a cuboid 240 cm by 120 cm by 100 cm. It is filled with boxes that are cuboids 60 cm by 30 cm by 20 cm.

How many boxes fit exactly into the carton?

16.

[Maximum mark: 4]

A fish tank is a cuboid 60 cm long, 40 cm wide and 30 cm high. It is $\frac{2}{3}$ full of water and sand. The ratio of the volume of water to the volume of sand is 3 : 1.

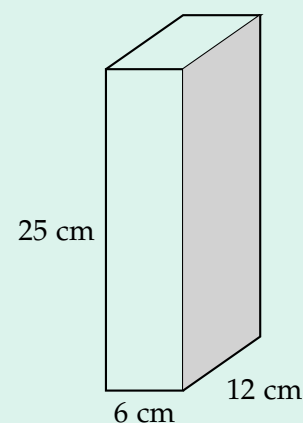
How many litres of water are in the tank? (1 litre = 1000 cm^3)

17.

[Maximum mark: 3]

A juice carton is a cuboid 6 cm wide, 12 cm deep and 25 cm tall. It stands on its 6 cm by 12 cm base and the juice is 10 cm deep.

The carton is closed and turned so that it stands on its 6 cm by 25 cm face (shaded). How deep is the juice now?



Homework answers E Volume – problem solving

14. $V = \frac{1}{2} \times x \times 12 \times 25 = 150x$. $150x = 1650 \Rightarrow x = \boxed{11 \text{ cm}}$

15. $240 \div 60 = 4$, $120 \div 30 = 4$, $100 \div 20 = 5$. Boxes = $4 \times 4 \times 5 = \boxed{80}$
(Check: $2\,880\,000 \div 36\,000 = 80 \checkmark$)

16. Tank = $60 \times 40 \times 30 = 72\,000 \text{ cm}^3$. $\frac{2}{3}$ full: $48\,000 \text{ cm}^3$.
Water is $\frac{3}{4}$ of this (3 parts out of 4): $48\,000 \times \frac{3}{4} = 36\,000 \text{ cm}^3 = \boxed{36 \text{ litres}}$

17. Juice = $6 \times 12 \times 10 = 720 \text{ cm}^3$. New base = $6 \times 25 = 150 \text{ cm}^2$.
Depth = $720 \div 150 = \boxed{4.8 \text{ cm}}$

Homework E Volume in real life

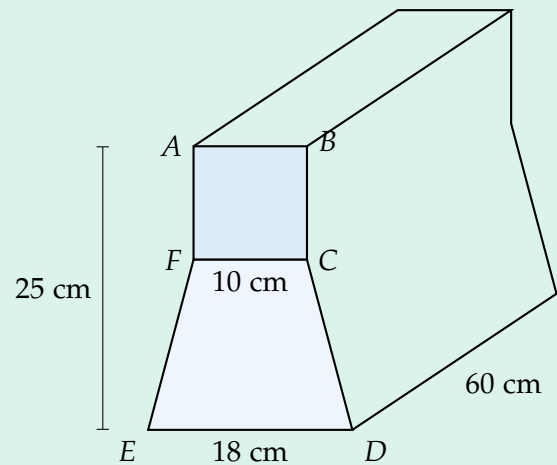
Section 4.5

18.

[Maximum mark: 4]

The cross-section of this prism is made of a square $ABCF$ of side 10 cm on top of a trapezium $FCDE$. $ED = 18$ cm and the total height of the cross-section is 25 cm. The prism is 60 cm long.

Work out the volume of the prism.



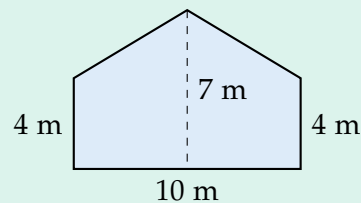
19.

[Maximum mark: 4]

The front face of a barn is a pentagon. It is 10 m wide, the side walls are 4 m high and the top of the roof is 7 m high.

a) Work out the area of the front face.

b) The barn is a prism 20 m long. Work out its volume.



20.

[Maximum mark: 3]

A water channel is a prism with a rectangular cross-section of area 0.8 m^2 . It is full, and the water flows along it at 5 metres per minute.

How many cubic metres of water flow along the channel in 2 hours?

21.

[Maximum mark: 3]

A water tank is a cuboid 2 m long and 1.2 m wide. The water in it is 45 cm deep.

How many litres of water are in the tank? ($1 \text{ m}^3 = 1000$ litres)

Homework answers E Volume in real life

18. Square = $10 \times 10 = 100 \text{ cm}^2$. Trapezium height = $25 - 10 = 15 \text{ cm}$, area = $\frac{1}{2}(10 + 18) \times 15 = 210 \text{ cm}^2$.

Cross-section = 310 cm^2 . $V = 310 \times 60 = \boxed{18\,600 \text{ cm}^3}$

19. a) Rectangle $10 \times 4 = 40$; triangle $\frac{1}{2} \times 10 \times (7 - 4) = 15$. Area = $\boxed{55 \text{ m}^2}$

b) $V = 55 \times 20 = \boxed{1100 \text{ m}^3}$

20. Each minute, a 5 m length of water passes: $0.8 \times 5 = 4 \text{ m}^3$ per minute.

2 hours = 120 minutes: $4 \times 120 = \boxed{480 \text{ m}^3}$

21. $45 \text{ cm} = 0.45 \text{ m}$. $V = 2 \times 1.2 \times 0.45 = 1.08 \text{ m}^3$. $1.08 \times 1000 = \boxed{1080 \text{ litres}}$

F Converting measures of area and volume

Section 4.6

1.

[Maximum mark: 6]

Copy and complete.

a) $5 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ mm}^2$

b) $2.6 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

c) $4500 \text{ mm}^2 = \underline{\hspace{2cm}} \text{ cm}^2$

d) $38\,000 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ m}^2$

e) $3 \text{ km}^2 = \underline{\hspace{2cm}} \text{ m}^2$

f) $75\,000 \text{ m}^2 = \underline{\hspace{2cm}} \text{ hectares}$

2.

[Maximum mark: 4]

Copy and complete.

a) $7 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ mm}^3$

b) $3500 \text{ mm}^3 = \underline{\hspace{2cm}} \text{ cm}^3$

c) $2.4 \text{ m}^3 = \underline{\hspace{2cm}} \text{ cm}^3$

d) $450\,000 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ m}^3$

3.

[Maximum mark: 4]

Copy and complete.

a) $2500 \text{ cm}^3 = \underline{\hspace{2cm}} \text{ litres}$

b) $750 \text{ ml} = \underline{\hspace{2cm}} \text{ cm}^3$

c) $1.8 \text{ m}^3 = \underline{\hspace{2cm}} \text{ litres}$

d) $0.35 \text{ litres} = \underline{\hspace{2cm}} \text{ ml}$

4.

[Maximum mark: 2]

Jan says, "There are 100 cm in 1 m, so $1 \text{ m}^2 = 100 \text{ cm}^2$." Explain why Jan is wrong.

Answers – F Converting measures

worked solutions

1. a) $5 \times 100 = 500 \text{ mm}^2$

b) $2.6 \times 10\,000 = 26\,000 \text{ cm}^2$

c) $4500 \div 100 = 45 \text{ cm}^2$

d) $38\,000 \div 10\,000 = 3.8 \text{ m}^2$

e) $3 \times 1\,000\,000 = 3\,000\,000 \text{ m}^2$

f) $75\,000 \div 10\,000 = 7.5 \text{ hectares}$

2. a) $7 \times 1000 = 7000 \text{ mm}^3$

b) $3500 \div 1000 = 3.5 \text{ cm}^3$

c) $2.4 \times 1\,000\,000 = 2\,400\,000 \text{ cm}^3$

d) $450\,000 \div 1\,000\,000 = 0.45 \text{ m}^3$

3. a) $2500 \div 1000 = 2.5 \text{ litres}$

b) 750 cm^3 ($1 \text{ ml} = 1 \text{ cm}^3$)

c) $1.8 \times 1000 = 1800 \text{ litres}$

d) $0.35 \times 1000 = 350 \text{ ml}$

4. A 1 m by 1 m square is 100 cm by 100 cm, so its area is $100 \times 100 = 10\,000 \text{ cm}^2$, not 100 cm^2 . Jan has used the length conversion factor instead of *squaring* it.

F Converting measures – part 2

Section 4.6

5. [Maximum mark: 2]

A rectangular field measures 400 m by 250 m. What is its area in hectares?

6. [Maximum mark: 3]

A rectangular rug is 2.5 m long and 40 cm wide. Work out its area in **a) m²** **b) cm²**.

7. [Maximum mark: 3]

A water tank is a cuboid 1.5 m by 0.8 m by 0.6 m. How many litres of water does it hold when full?

8. [Maximum mark: 3]

A juice carton is a cuboid 20 cm by 6 cm by 10 cm.

How many full 250 ml glasses can be poured from a full carton?

9. [Maximum mark: 4]

Healthy soil should have at least 5000 earthworms per cubic metre.

Fred digs out a sample of soil measuring 25 cm by 20 cm by 20 cm and finds 45 earthworms.

Is his soil healthy? Show your working.

Answers – F Converting measures – part 2 worked solutions

5. $400 \times 250 = 100\,000 \text{ m}^2$. $100\,000 \div 10\,000 =$ 10 hectares

6. a) $40 \text{ cm} = 0.4 \text{ m}$, $2.5 \times 0.4 =$ 1 m² b) $250 \times 40 =$ 10 000 cm² (same as $1 \times 10\,000$ ✓)

7. $V = 1.5 \times 0.8 \times 0.6 = 0.72 \text{ m}^3$. $0.72 \times 1000 =$ 720 litres

8. $V = 20 \times 6 \times 10 = 1200 \text{ cm}^3 = 1200 \text{ ml}$. $1200 \div 250 = 4.8$, so 4 full glasses (round down).

9. Sample = $25 \times 20 \times 20 = 10\,000 \text{ cm}^3$. $10\,000 \div 1\,000\,000 = 0.01 \text{ m}^3$.

Worms per m³ = $45 \div 0.01 = 4500$.

$4500 < 5000$, so the soil is **not healthy** (it needs at least 50 worms in this sample).

Homework F Converting measures

Section 4.6

1.

[Maximum mark: 6]

Copy and complete.

a) $9 \text{ cm}^2 = \text{_____ mm}^2$

b) $0.75 \text{ m}^2 = \text{_____ cm}^2$

c) $63\,000 \text{ cm}^2 = \text{_____ m}^2$

d) $250 \text{ mm}^2 = \text{_____ cm}^2$

e) $4.2 \text{ hectares} = \text{_____ m}^2$

f) $0.6 \text{ km}^2 = \text{_____ hectares}$

2.

[Maximum mark: 4]

Copy and complete.

a) $0.5 \text{ cm}^3 = \text{_____ mm}^3$

b) $12\,000 \text{ mm}^3 = \text{_____ cm}^3$

c) $0.08 \text{ m}^3 = \text{_____ cm}^3$

d) $3\,500\,000 \text{ cm}^3 = \text{_____ m}^3$

3.

[Maximum mark: 4]

Copy and complete.

a) $4.5 \text{ litres} = \text{_____ cm}^3$

b) $600 \text{ cm}^3 = \text{_____ ml}$

c) $2.5 \text{ m}^3 = \text{_____ litres}$

d) $3200 \text{ ml} = \text{_____ litres}$

4.

[Maximum mark: 2]

Lily says, “ 3.5 m^2 is the same as 350 cm^2 .” Is she right? Explain.

Homework answers F Converting measures

1. a) $9 \times 100 = 900 \text{ mm}^2$

b) $0.75 \times 10\,000 = 7500 \text{ cm}^2$

c) $63\,000 \div 10\,000 = 6.3 \text{ m}^2$

d) $250 \div 100 = 2.5 \text{ cm}^2$

e) $4.2 \times 10\,000 = 42\,000 \text{ m}^2$

f) $0.6 \times 100 = 60 \text{ hectares}$ ($1 \text{ km}^2 = 100 \text{ ha}$)

2. a) $0.5 \times 1000 = 500 \text{ mm}^3$

b) $12\,000 \div 1000 = 12 \text{ cm}^3$

c) $0.08 \times 1\,000\,000 = 80\,000 \text{ cm}^3$

d) $3\,500\,000 \div 1\,000\,000 = 3.5 \text{ m}^3$

3. a) 4500 cm^3

b) 600 ml

c) $2.5 \times 1000 = 2500 \text{ litres}$

d) $3200 \div 1000 =$

3.2 litres

4. **No.** $1 \text{ m}^2 = 100 \times 100 = 10\,000 \text{ cm}^2$, so $3.5 \text{ m}^2 = 3.5 \times 10\,000 = 35\,000 \text{ cm}^2$. Lily multiplied by 100 instead of 100^2 .

Homework F Converting measures – problems Section 4.6

5. [Maximum mark: 3]

A rug is 3.2 m long and 85 cm wide. Work out its area in **a)** cm^2 **b)** m^2 .

6. [Maximum mark: 2]

A park is a rectangle 0.45 km by 0.32 km. What is its area in hectares?

7. [Maximum mark: 3]

A cube-shaped water tank has edges of 1.5 m. How many litres does it hold when full?

8. [Maximum mark: 3]

A medicine bottle holds 0.3 litres. One dose is 5 ml and Ben takes a dose 3 times a day.

a) How many doses are in the bottle? **b)** How many days will the bottle last?

9. [Maximum mark: 2]

How many 1 cm cubes would fit inside a cube with edges of 1 metre?

Homework answers F Converting measures – problems

5. a) $320 \times 85 = \boxed{27\,200 \text{ cm}^2}$ b) $3.2 \times 0.85 = \boxed{2.72 \text{ m}^2}$ (check: $27\,200 \div 10\,000 = 2.72$ ✓)

6. $450 \text{ m} \times 320 \text{ m} = 144\,000 \text{ m}^2$. $144\,000 \div 10\,000 = \boxed{14.4 \text{ hectares}}$

7. $V = 1.5^3 = 3.375 \text{ m}^3$. $3.375 \times 1000 = \boxed{3375 \text{ litres}}$

8. a) $0.3 \text{ litres} = 300 \text{ ml}$. $300 \div 5 = \boxed{60 \text{ doses}}$ b) $60 \div 3 = \boxed{20 \text{ days}}$

9. $1 \text{ m} = 100 \text{ cm}$, so $100 \times 100 \times 100 = \boxed{1\,000\,000 \text{ cubes}}$ ($1 \text{ m}^3 = 1\,000\,000 \text{ cm}^3$).

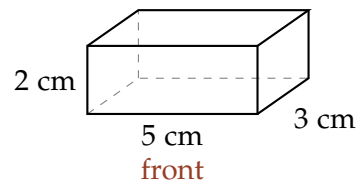
G Plans and elevations

Section 4.7

1.

[Maximum mark: 3]

Sketch the plan, front elevation and side elevation of this cuboid. Label the lengths.



2.

[Maximum mark: 4]

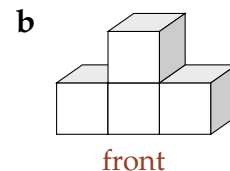
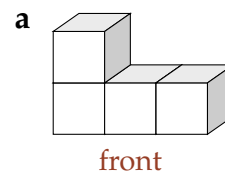
Each row shows the plan, front elevation and side elevation of a solid. Name each solid.

	Plan	Front	Side	
a				_____
b				_____
c				_____
d				_____

3.

[Maximum mark: 4]

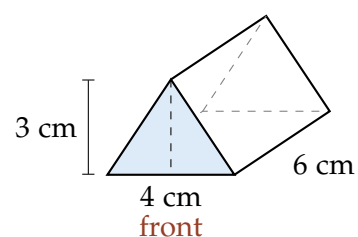
These solids are made from centimetre cubes. For each one, draw the plan, front elevation and side elevation on squared paper.



4.

[Maximum mark: 3]

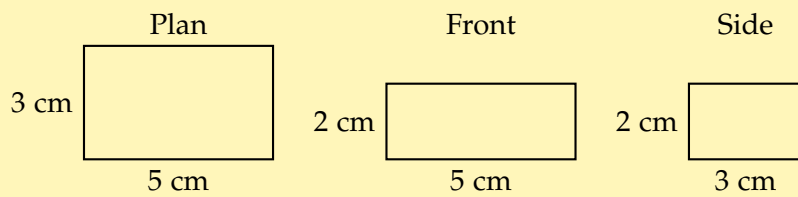
A triangular prism has a triangular front face (base 4 cm, height 3 cm) and is 6 cm long. Sketch its plan, front elevation and side elevation.



Answers – G Plans and elevations

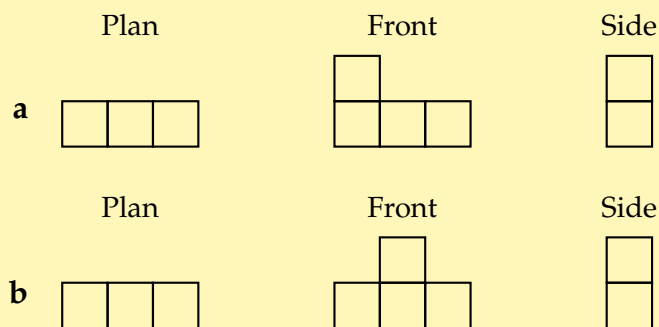
worked solutions

1.



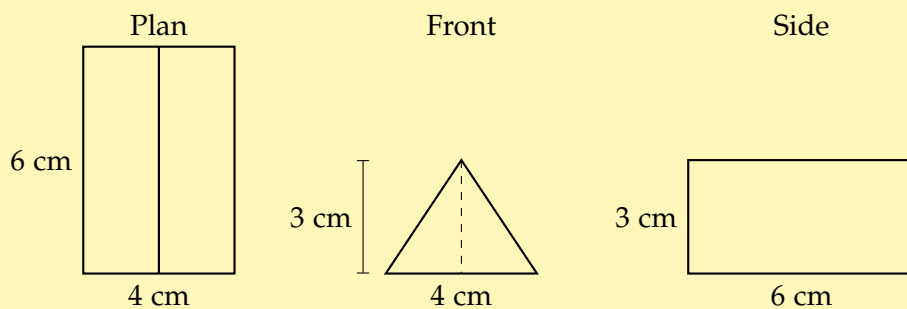
2. a) Cylinder b) Square-based pyramid c) Cone d) Triangular prism

3.



The plans and side elevations of **a** and **b** are the same – only the front elevation shows the difference.

4.



The line down the middle of the plan is the top edge (ridge) of the prism.

Homework G From views to solids

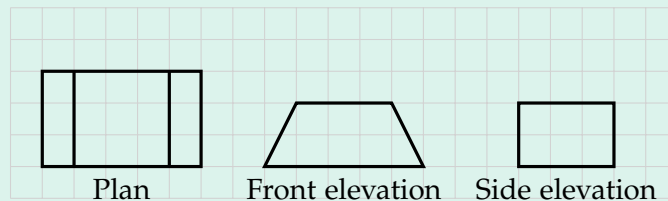
Section 4.7

Each square on the grids is 1 cm.

1.

[Maximum mark: 2]

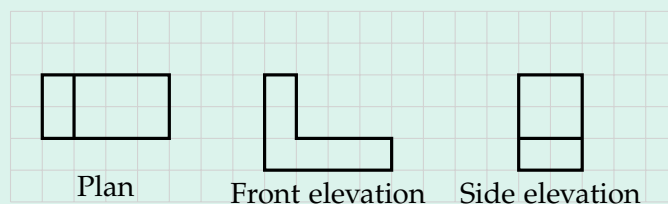
The plan, front elevation and side elevation of a solid are drawn on a centimetre grid. Sketch the solid and write its dimensions on your sketch. What is the name of the solid?



2.

[Maximum mark: 2]

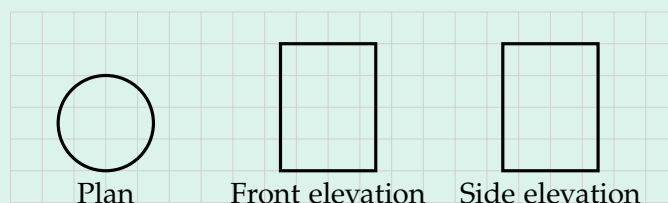
Sketch the solid shown by these views. Give its dimensions.



3.

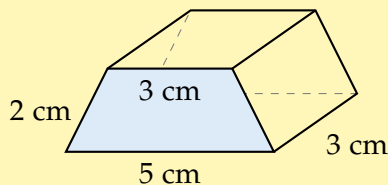
[Maximum mark: 2]

Sketch the solid shown by these views. Give its dimensions.

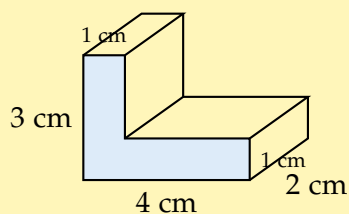


Homework answers G From views to solids

1. A **trapezium (trapezoidal) prism**. Front face: trapezium with bottom 5 cm, top 3 cm, height 2 cm. Length (depth) 3 cm.



2. An **L-shaped prism**: front is an L, 4 cm wide and 3 cm tall; the upright part is 1 cm wide and the base part is 1 cm high. Depth 2 cm.



3. A **cylinder** with diameter 3 cm (radius 1.5 cm) and height 4 cm.

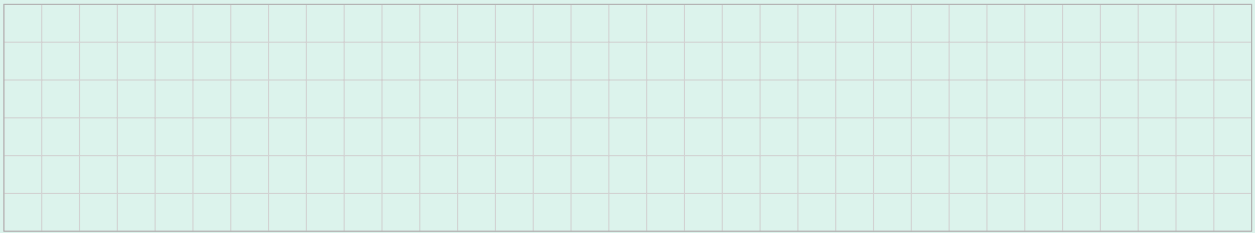
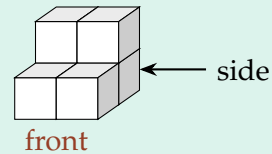
Homework G Drawing plans and elevations

Section 4.7

4.

[Maximum mark: 3]

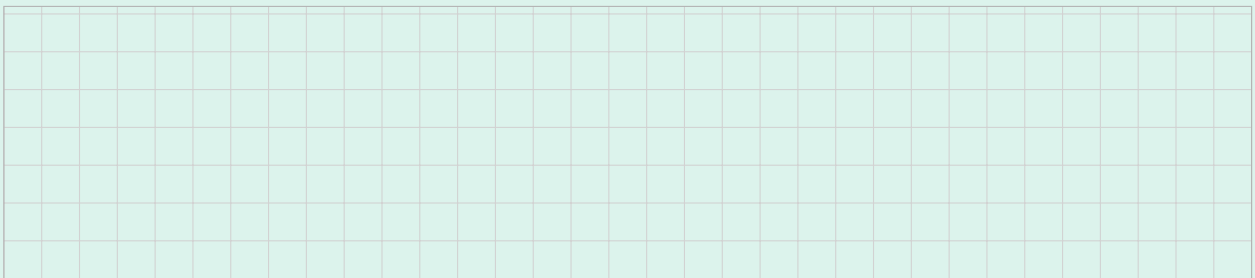
This solid is made from centimetre cubes.
Draw its **plan** and its **side elevation** (from the direction of the arrow) on the grid.



5.

[Maximum mark: 3]

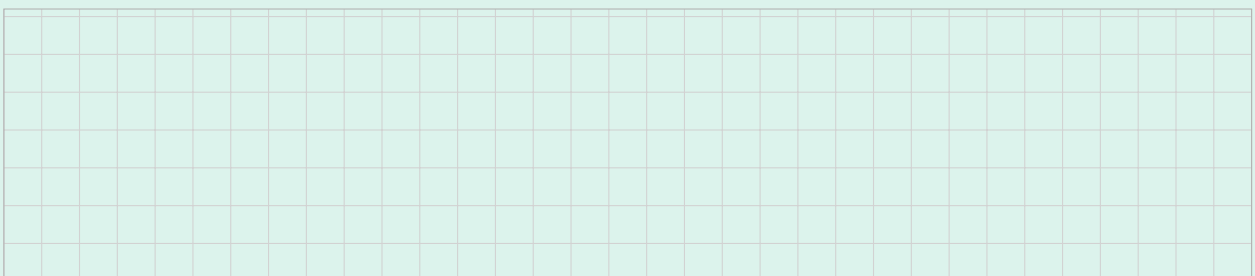
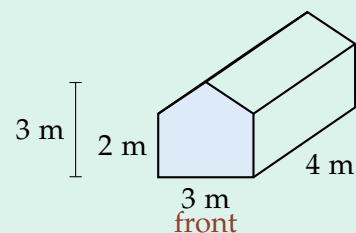
A cone has radius 2 cm and perpendicular height 6 cm. Draw its **plan** and its **side elevation**.



6.

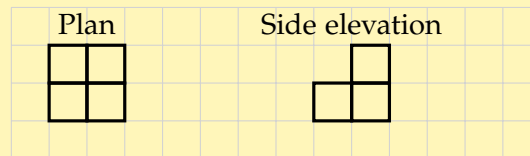
[Maximum mark: 4]

The diagram shows a prism. Using a scale of 1 cm to 1 m (2 small squares = 1 m), draw the **front elevation** and the **side elevation**.



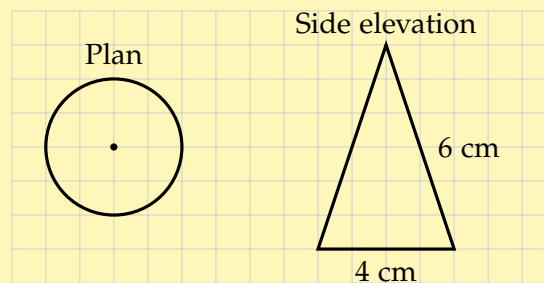
Homework answers G Drawing plans and elevations

4.



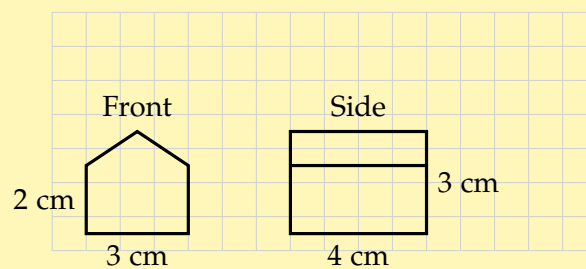
The plan is a 2 by 2 square. From the side, the front column is 1 cube high and the back column is 2 cubes high. (A mirror-image side view is also accepted.)

5.



Plan: a circle of radius 2 cm (with a dot for the vertex). Side elevation: isosceles triangle, base 4 cm, height 6 cm.

6.



Front: pentagon 3 cm wide, walls 2 cm, top 3 cm high. Side: rectangle 4 cm by 3 cm with a line 2 cm up (where the roof meets the walls).

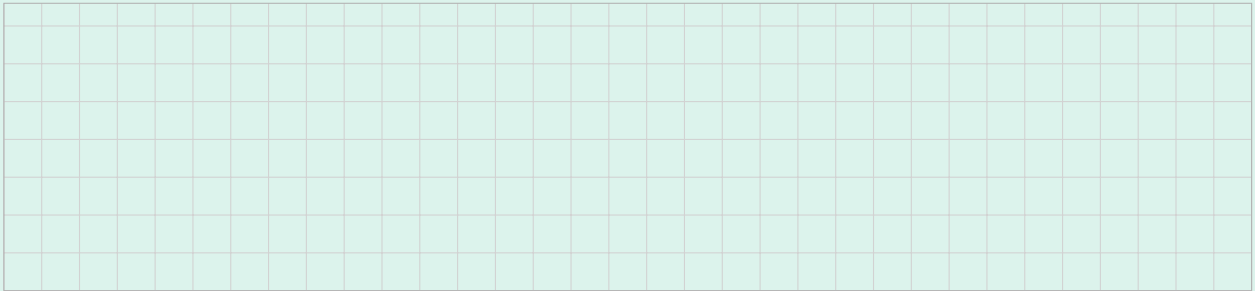
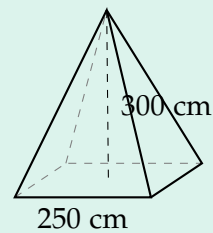
Homework G Plans and elevations – cubes

Section 4.7

7.

[Maximum mark: 4]

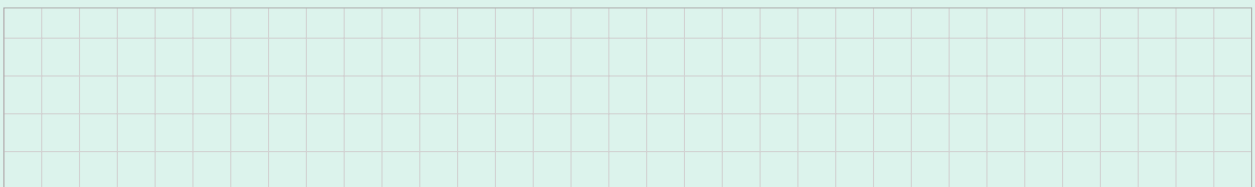
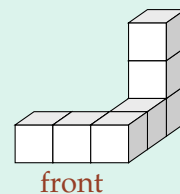
A square-based pyramid has a base of side 250 cm and a perpendicular height of 300 cm. Using a scale of **2 cm to 1 m**, draw the **plan** and the **front elevation**.



8.

[Maximum mark: 2]

This solid is made from seven centimetre cubes. Draw the **plan** of the solid on the grid.

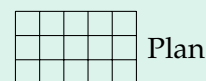


9.

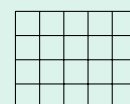
[Maximum mark: 2]

A solid cuboid is made from centimetre cubes. Its plan, front elevation and side elevation are shown.

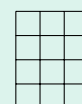
How many cubes were used?



Plan



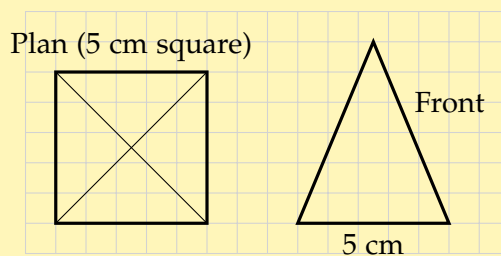
Front



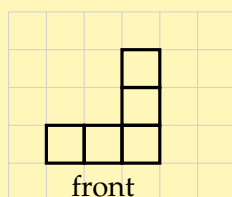
Side

Homework answers G Plans and elevations – cubes

7. Scale 2 cm to 1 m: base 2.5 m $\rightarrow$ **5 cm**; height 3 m $\rightarrow$ **6 cm**.



8.



Plan: an L of 5 squares (the tower of 3 cubes only shows as one square from above).

9. Length 5, width 3 (from the plan) and height 4 (from the front). $5 \times 3 \times 4 =$ 60 cubes

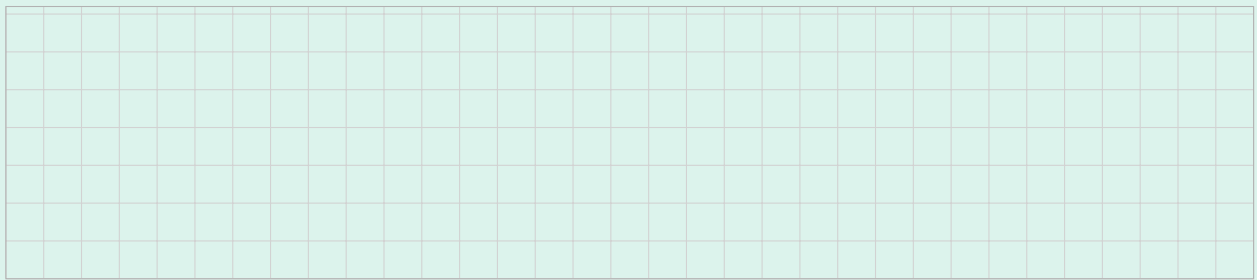
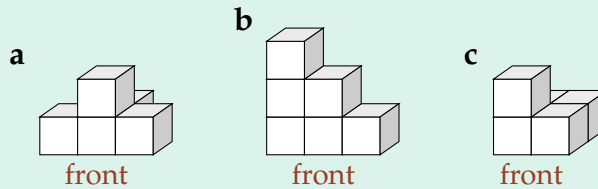
Homework G Front elevations and plans

Section 4.7

10.

[Maximum mark: 6]

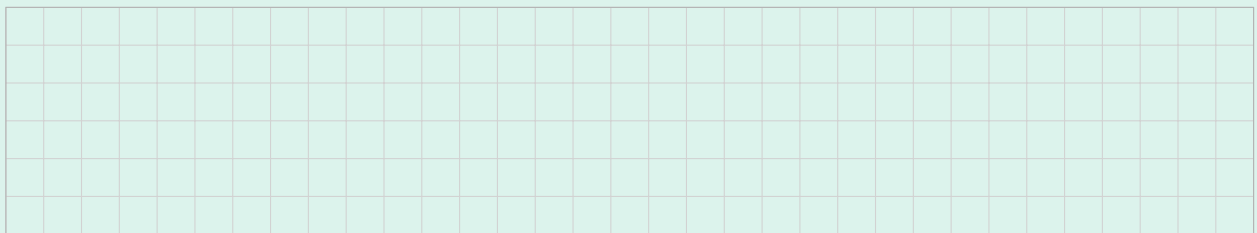
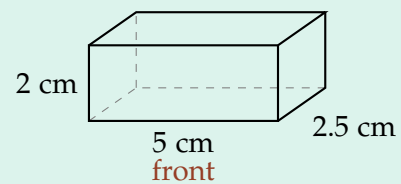
Each solid is made from centimetre cubes. For each one, draw **(i)** the front elevation and **(ii)** the plan.



11.

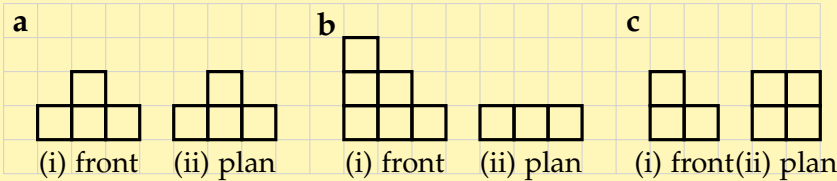
[Maximum mark: 4]

Here is a cuboid. Draw **a)** the front elevation
b) the plan, full size.



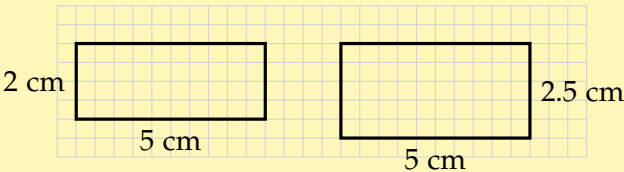
Homework answers G Front elevations and plans

10.



In each plan the bottom row of squares is the front of the solid.

11. a) Front elevation: rectangle 5 cm by 2 cm. b) Plan: rectangle 5 cm by 2.5 cm.



Homework G Working backwards

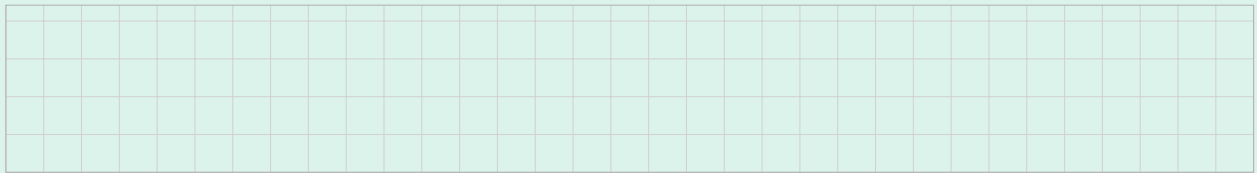
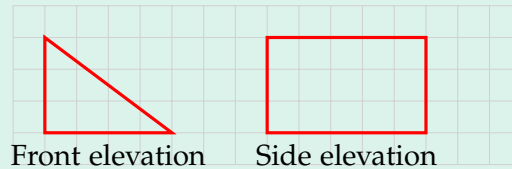
Section 4.7

12.

[Maximum mark: 4]

Here are the front and side elevations of a solid.

a) Draw the plan on the grid. b) Sketch the solid and name it.

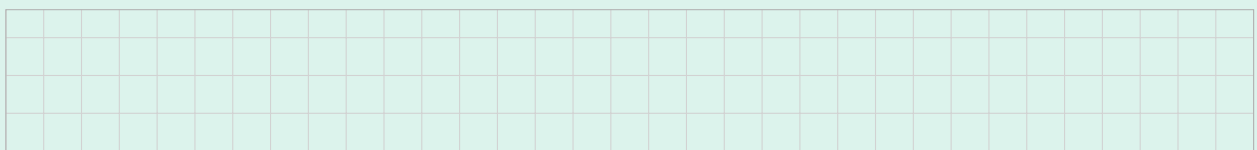
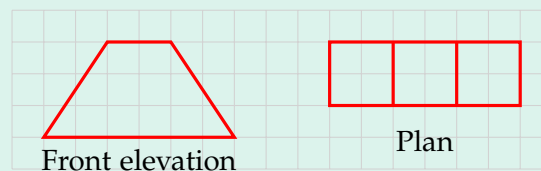


13.

[Maximum mark: 4]

Here are the front elevation and the plan of a solid.

a) Draw the side elevation on the grid. b) Sketch the solid.



14.

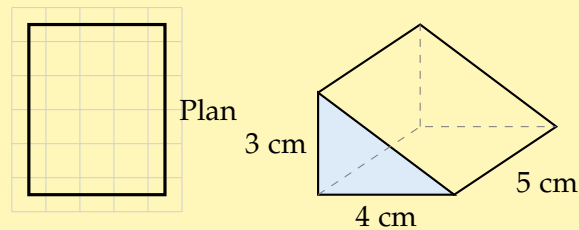
[Maximum mark: 4]

Name each solid.

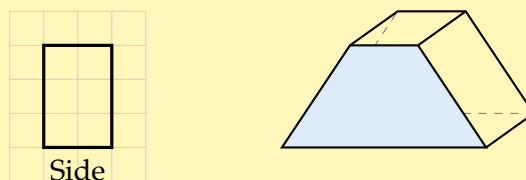
- a) Front elevation: rectangle. Side elevation: rectangle. Plan: circle.
- b) Front elevation: triangle. Side elevation: rectangle. Plan: rectangle.
- c) Front elevation: square. Side elevation: square. Plan: square.
- d) Front elevation: triangle. Side elevation: triangle. Plan: triangle with lines from each corner to the middle.

Homework answers G Working backwards

- 12. a)** The plan is a rectangle 4 cm wide (from the front) and 5 cm deep (from the side).
b) A **triangular prism** lying on one of its rectangular faces: a right-angled triangle (base 4 cm, height 3 cm) at the front, 5 cm long.



- 13. a)** Side elevation: a rectangle 2 cm wide (the depth, from the plan) and 3 cm tall.
b) A **trapezium prism**: front is a trapezium (bottom 6 cm, top 2 cm, height 3 cm), 2 cm deep.



- 14. a)** Cylinder (standing up) **b)** Triangular prism **c)** Cube **d)** Tetrahedron (triangular-based pyramid)

H Mixed problem solving

Section 4.8

1. [Maximum mark: 3]

A medicine bottle holds 0.2 litres. The label says, “Take two 5 ml spoonfuls three times a day.”

Priya has to take the medicine for 7 days. Is there enough medicine in the bottle? Explain your answer.

2. [Maximum mark: 2]

A lorry can carry a maximum load of 3.5 tonnes. Bags of cement weigh 25 kg each.

What is the greatest number of bags the lorry can carry?

3. [Maximum mark: 4]

A swimming pool is a cuboid 25 m long, 10 m wide and 2 m deep.

a) How many litres of water does it hold when full?

b) A pump fills the pool at 250 litres per minute. How long does it take to fill? Give your answer in hours and minutes.

4. [Maximum mark: 3]

A present is in a cube-shaped box with edges of 15 cm. Lily has a sheet of wrapping paper 30 cm by 40 cm.

Is the area of the paper enough to cover the whole box? Show how you decide.

Answers – H Mixed problem solving

worked solutions

1. Per day: $2 \times 5 \times 3 = 30 \text{ ml}$. For 7 days: $30 \times 7 = 210 \text{ ml}$.
Bottle = 0.2 litres = 200 ml. $200 < 210$, so **no, there is not enough** – it is 10 ml short.
2. 3.5 tonnes = 3500 kg. $3500 \div 25 = \boxed{140 \text{ bags}}$
3. a) $V = 25 \times 10 \times 2 = 500 \text{ m}^3$. $500 \times 1000 = \boxed{500\,000 \text{ litres}}$
b) $500\,000 \div 250 = 2000 \text{ minutes}$. $2000 \div 60 = 33 \text{ remainder } 20$, so $\boxed{33 \text{ hours } 20 \text{ minutes}}$
4. Box: $SA = 6 \times 15 \times 15 = 6 \times 225 = 1350 \text{ cm}^2$. Paper: $30 \times 40 = 1200 \text{ cm}^2$.
 $1200 < 1350$, so **no** – the paper is 150 cm^2 too small (even before allowing for overlaps).

H Mixed problem solving – part 2

Section 4.8

5.

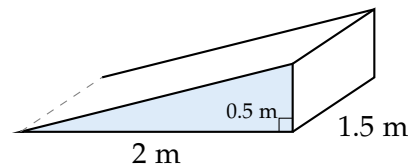
[Maximum mark: 3]

A container is a cuboid 20 cm by 10 cm by 15 cm and is full of water. All the water is poured into an empty tank with a rectangular base 25 cm by 12 cm. How deep is the water in the tank?

6.

[Maximum mark: 4]

A concrete ramp is a triangular prism. Its cross-section is a right-angled triangle 0.5 m high and 2 m long. The ramp is 1.5 m wide. Concrete costs £90 per m^3 . How much does the concrete for the ramp cost?



7.

[Maximum mark: 3]

A cube has a volume of 27 cm^3 . Work out its surface area.

8.

[Maximum mark: 5]

Cuboid A measures 8 cm by 6 cm by 5 cm.

Cuboid B has the **same volume** as A. Its base is 10 cm by 4 cm.

a) Find the height of cuboid B. **b)** Which cuboid has the larger surface area, and by how much?

Answers – H Mixed problems – part 2

worked solutions

5. Volume of water = $20 \times 10 \times 15 = 3000 \text{ cm}^3$. Base of tank = $25 \times 12 = 300 \text{ cm}^2$.

$$\text{Depth} = 3000 \div 300 = \boxed{10 \text{ cm}}$$

6. Cross-section = $\frac{1}{2} \times 2 \times 0.5 = 0.5 \text{ m}^2$. $V = 0.5 \times 1.5 = 0.75 \text{ m}^3$.

$$\text{Cost} = 0.75 \times \pounds 90 = \boxed{\pounds 67.50}$$

7. Edge = $\sqrt[3]{27} = 3 \text{ cm}$. $SA = 6 \times 3 \times 3 = \boxed{54 \text{ cm}^2}$

8. a) Volume of A = $8 \times 6 \times 5 = 240 \text{ cm}^3$. $10 \times 4 \times h = 240 \Rightarrow h = \boxed{6 \text{ cm}}$

$$\text{b) } SA_A = 2(8 \times 6 + 8 \times 5 + 6 \times 5) = 2(48 + 40 + 30) = 236 \text{ cm}^2.$$

$$SA_B = 2(10 \times 4 + 10 \times 6 + 4 \times 6) = 2(40 + 60 + 24) = 248 \text{ cm}^2.$$

Cuboid B has the larger surface area, by $248 - 236 = \boxed{12 \text{ cm}^2}$.

Homework H Mixed problem solving

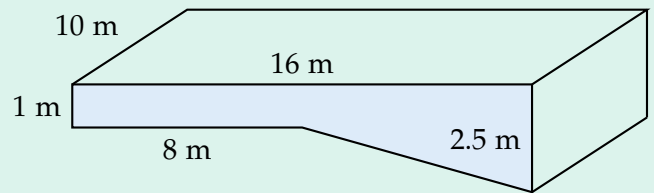
Section 4.8

1.

[Maximum mark: 5]

A swimming pool is a prism. It is 16 m long and 10 m wide. It is 1 m deep for the first 8 m, then the floor slopes down to 2.5 m deep at the far end.

The pool is filled at 6 litres per second. Jeremy has 10 hours to fill it. Will he fill it in time? Show your working. ($1 \text{ m}^3 = 1000$ litres)



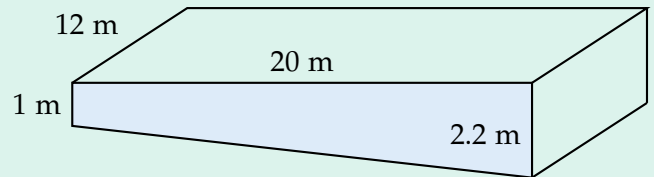
2.

[Maximum mark: 6]

A swimming pool is full of water. It is a prism with a trapezium cross-section: 20 m long, 1 m deep at one end and 2.2 m deep at the other. The pool is 12 m wide.

Water is pumped out at a constant rate. The water level drops by 4 cm in the first 20 minutes.

How long will it take to empty the pool? Give your answer in hours and minutes.



Homework answers H Mixed problem solving

1. Cross-section = rectangle 16×1 + triangle under the sloping part $\frac{1}{2} \times 8 \times 1.5 = 16 + 6 = 22 \text{ m}^2$.

Volume = $22 \times 10 = 220 \text{ m}^3 = 220\,000$ litres.

In 10 hours: $10 \times 60 \times 60 = 36\,000 \text{ s}$, $36\,000 \times 6 = 216\,000$ litres.

$216\,000 < 220\,000$, so **no** – he will not quite fill it (he needs about 10 h 11 min).

2. Volume = $\frac{1}{2}(1 + 2.2) \times 20 \times 12 = 32 \times 12 = 384 \text{ m}^3$.

Top 4 cm layer = $20 \times 12 \times 0.04 = 9.6 \text{ m}^3$ in 20 minutes $\Rightarrow 0.48 \text{ m}^3$ per minute.

Time = $384 \div 0.48 = 800$ minutes = 13 hours 20 minutes

Homework H Mixed problem solving – part 2 Section 4.8

3. [Maximum mark: 4]

A rectangular patio is 4.5 m by 3.6 m. It is covered with square slabs of side 45 cm. Slabs are sold in packs of 10 for £32.

How much do the slabs cost?

4. [Maximum mark: 3]

A fish tank is a cuboid 50 cm long, 30 cm wide and 36 cm tall. A jug holds 1.5 litres.

How many full jugs are needed to fill the tank to 4 cm below the top?

5. [Maximum mark: 4]

A raised flower bed is a cuboid 2.4 m long, 1.2 m wide and 0.4 m deep. Soil is sold in 50 litre bags costing £4.50 each.

How much will it cost to fill the bed with soil?

6. [Maximum mark: 3]

Which holds more: a cube with edges of 12 cm, or a cuboid 20 cm by 9 cm by 10 cm? By how much?

Homework answers H Mixed problem solving – part 2

3. Along: $4.5 \div 0.45 = 10$ slabs; across: $3.6 \div 0.45 = 8$ slabs. $10 \times 8 = 80$ slabs = 8 packs.

$$\text{Cost} = 8 \times £32 = \boxed{£256}$$

4. Water depth = $36 - 4 = 32$ cm. $50 \times 30 \times 32 = 48\,000 \text{ cm}^3 = 48$ litres.

$$48 \div 1.5 = \boxed{32 \text{ jugs}}$$

5. $V = 2.4 \times 1.2 \times 0.4 = 1.152 \text{ m}^3 = 1152$ litres. $1152 \div 50 = 23.04$, so **24 bags** (round up).

$$\text{Cost} = 24 \times £4.50 = \boxed{£108}$$

6. Cube = $12^3 = 1728 \text{ cm}^3$. Cuboid = $20 \times 9 \times 10 = 1800 \text{ cm}^3$.

The **cuboid** holds more, by $1800 - 1728 = \boxed{72 \text{ cm}^3}$.

Well done! Go back to any section where you dropped marks, re-read the pink notes for it, and try those questions again.